Cloud Storage & Networking Basics Activity

Cloud Computing

Contents

Activity 1 - Create and configure an S3 bucket

1.1 Prerequisite

- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

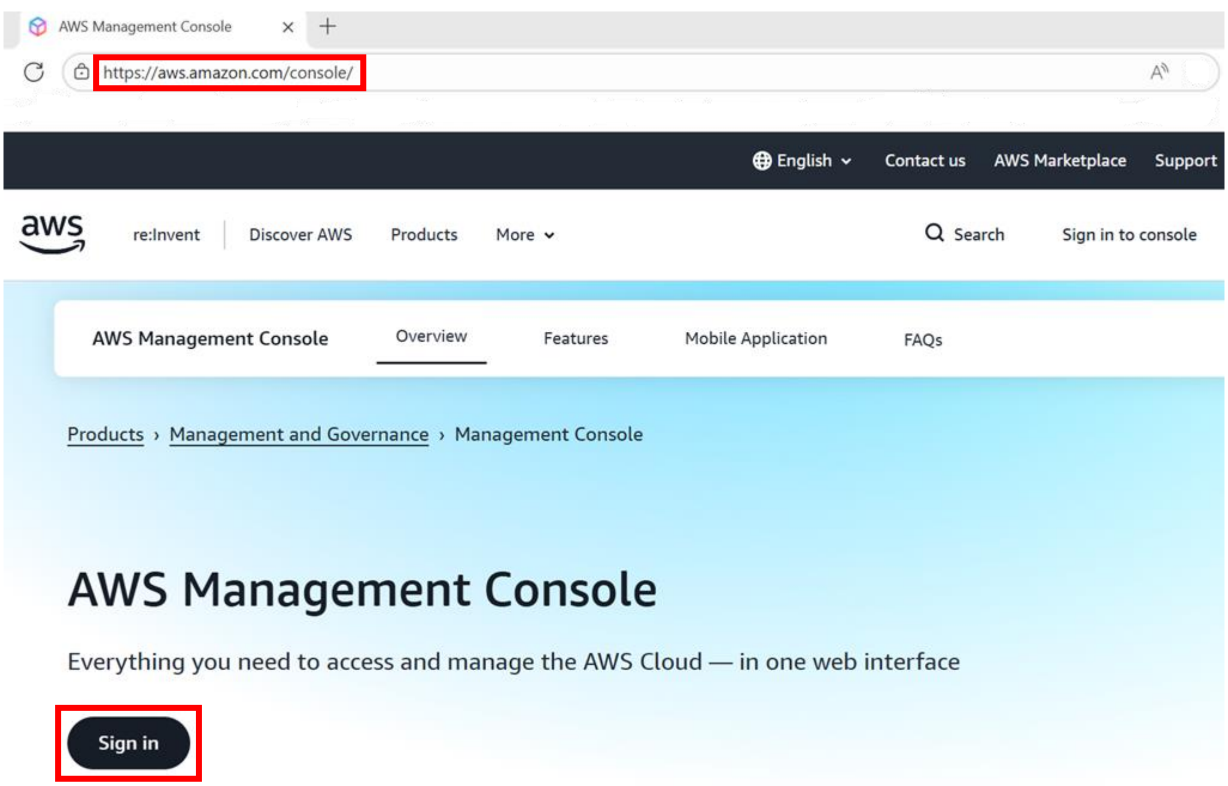
1.2 Scenario

Create and configure an S3 bucket

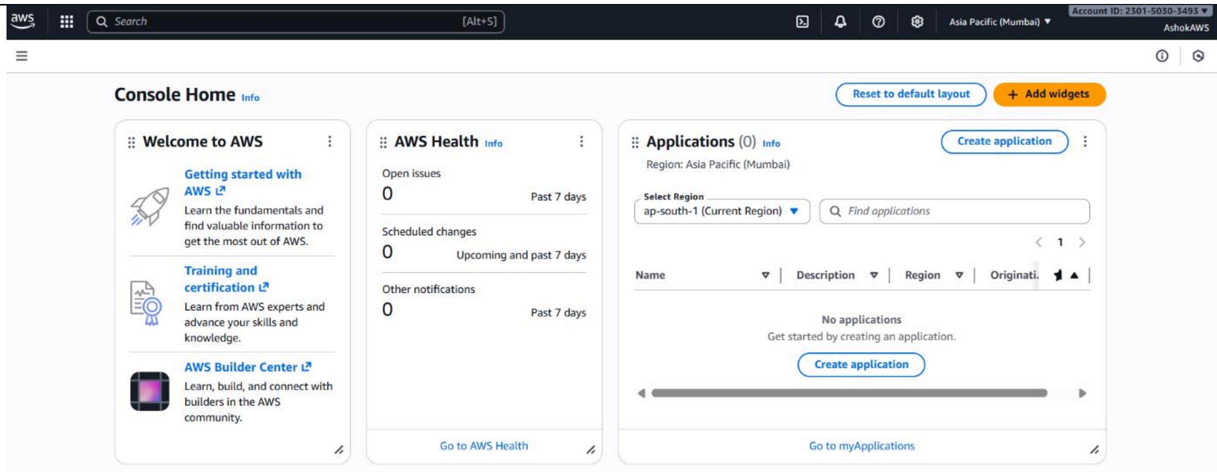
1.3 Steps

<table>
<tr>
<td>1</td>
<td>
Log in to AWS Console

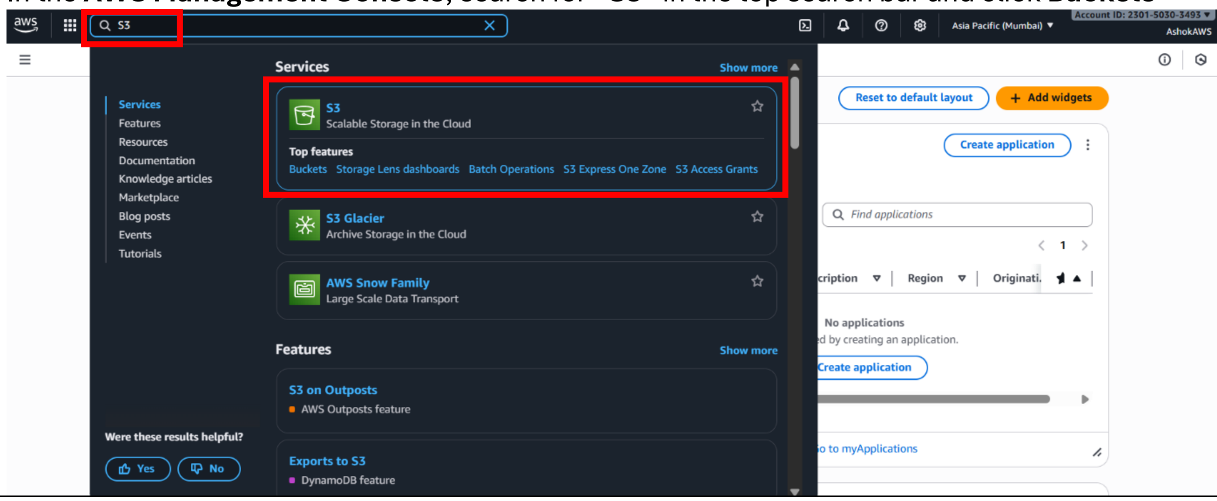
Visit https://aws.amazon.com/console
Sign in with your credentials



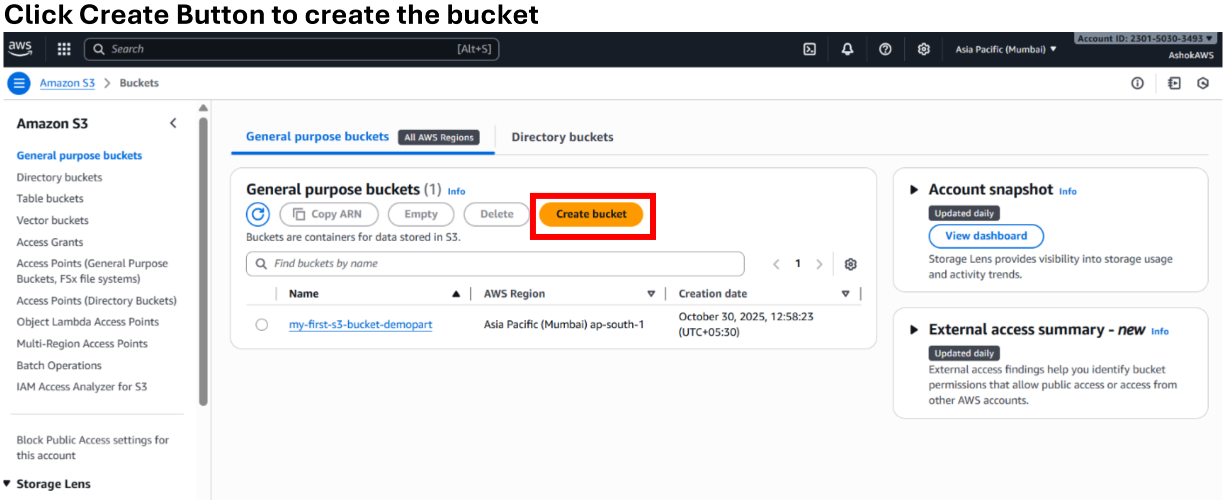
After Sign in, the following page will be displayed:
</td>
</tr>
</table>



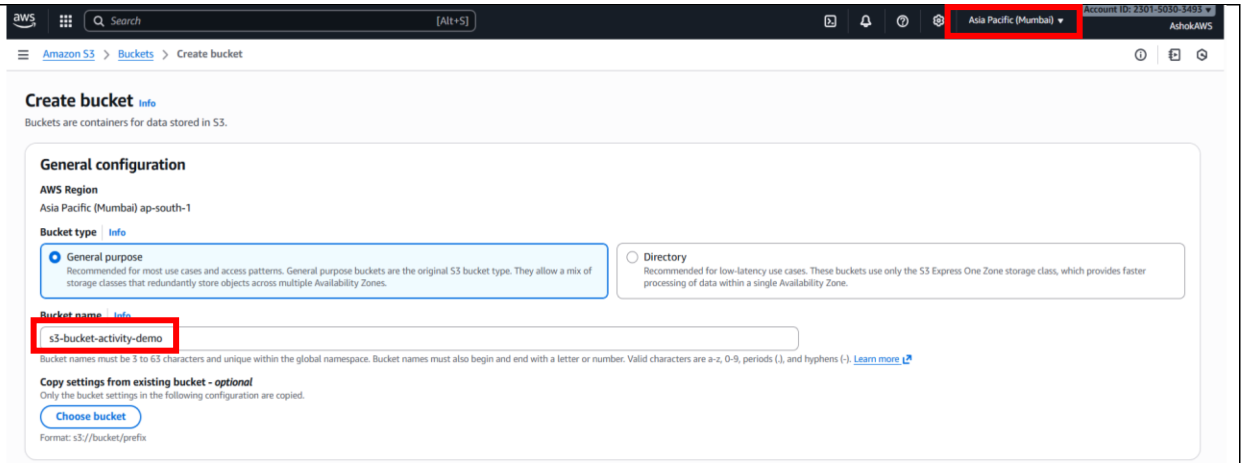
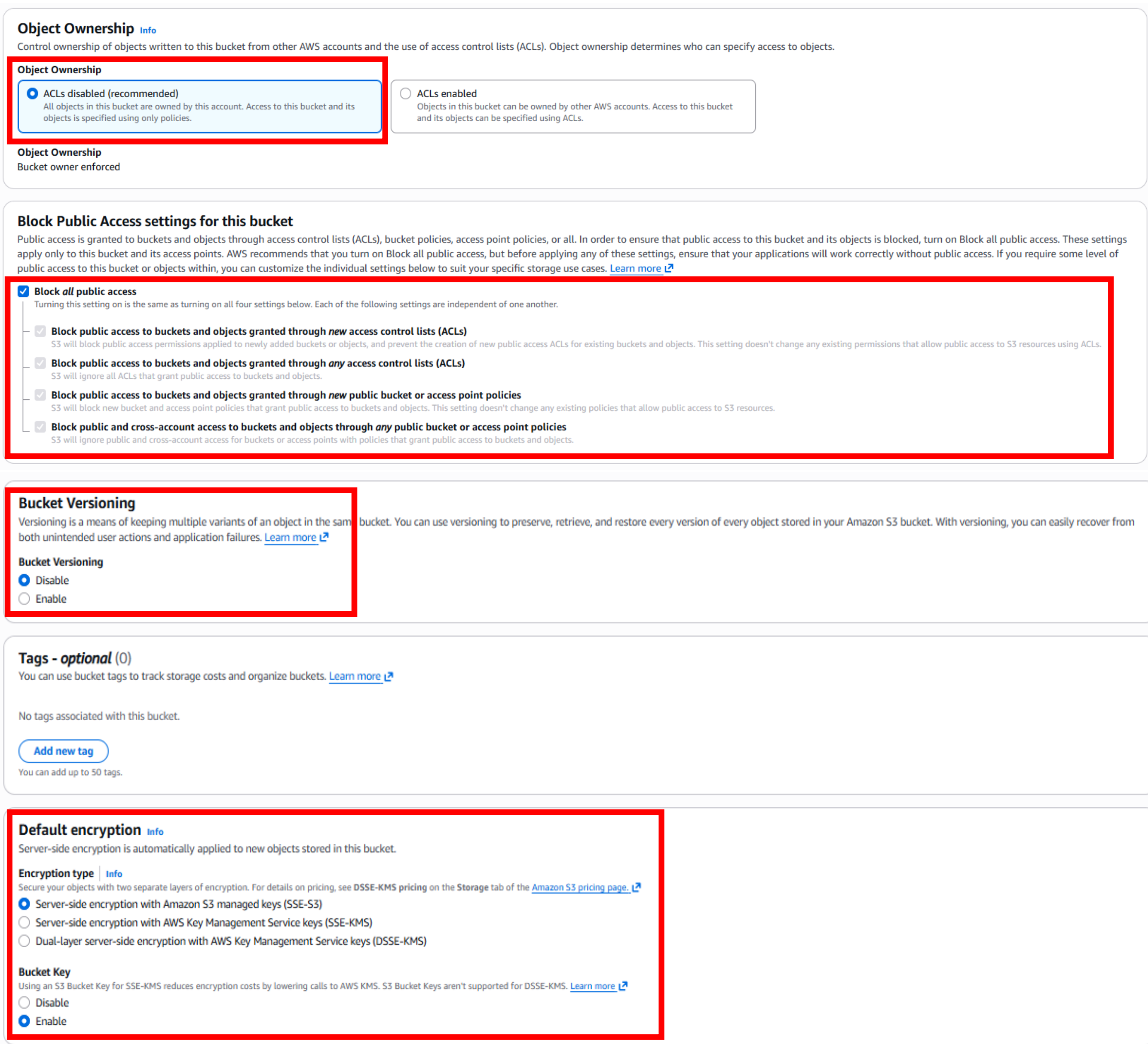
2 In the **AWS Management Console**, search for **“S3”** in the top search bar and click **Buckets**

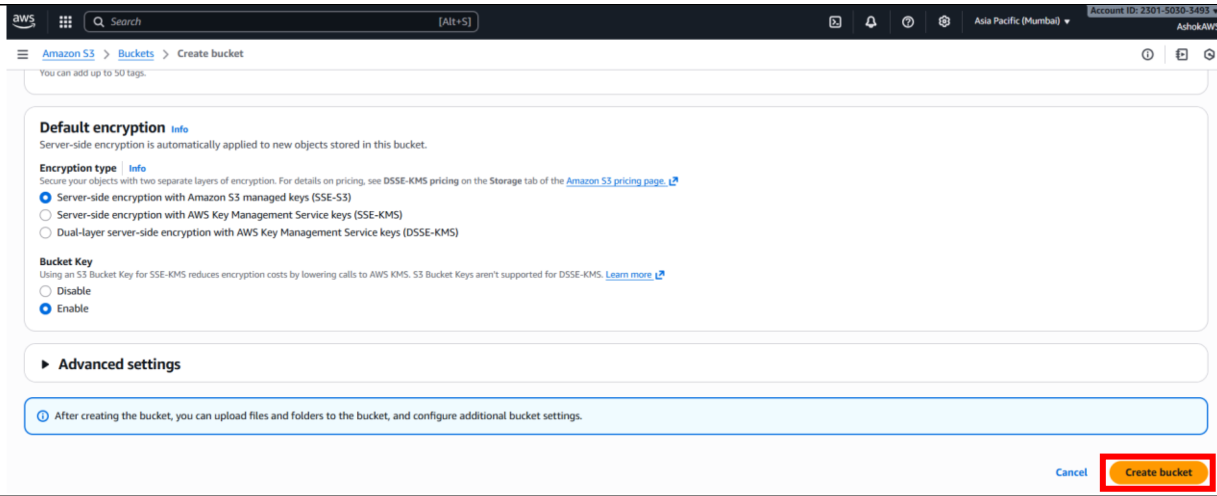
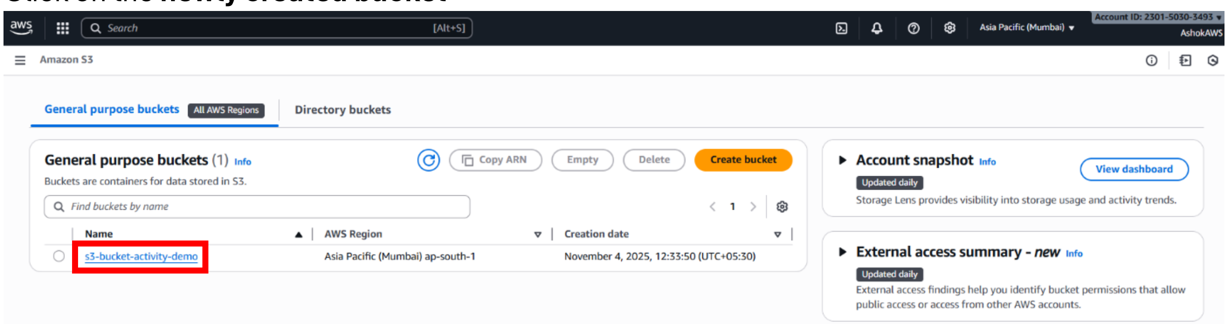


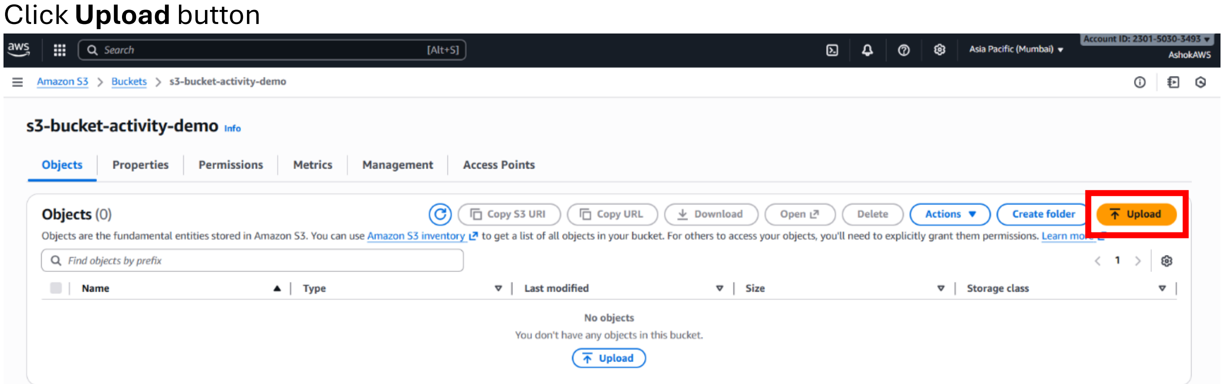
3 **Click Create Button to create the bucket**



4 Enter a **unique bucket name** and choose the **AWS Region** closest to your users or resources.

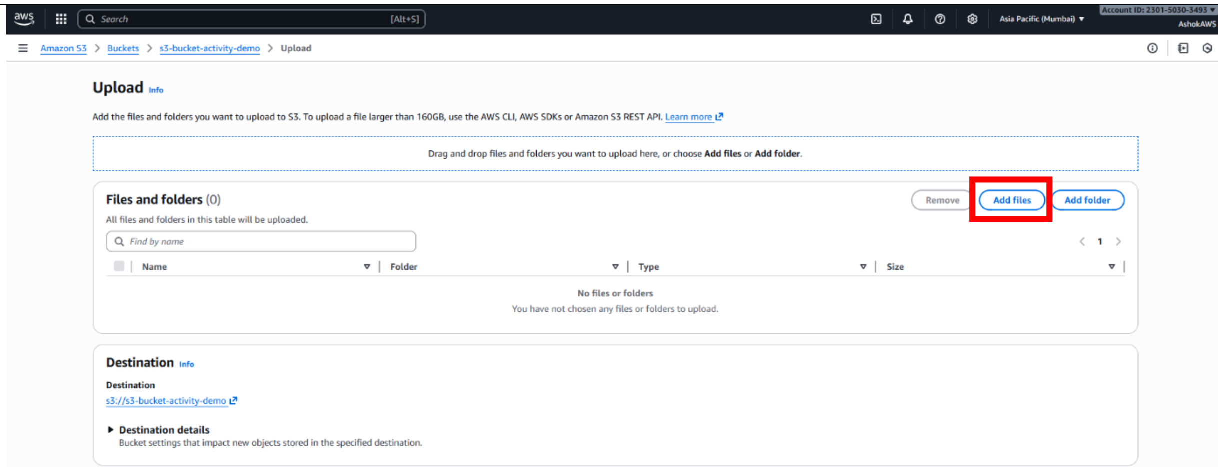
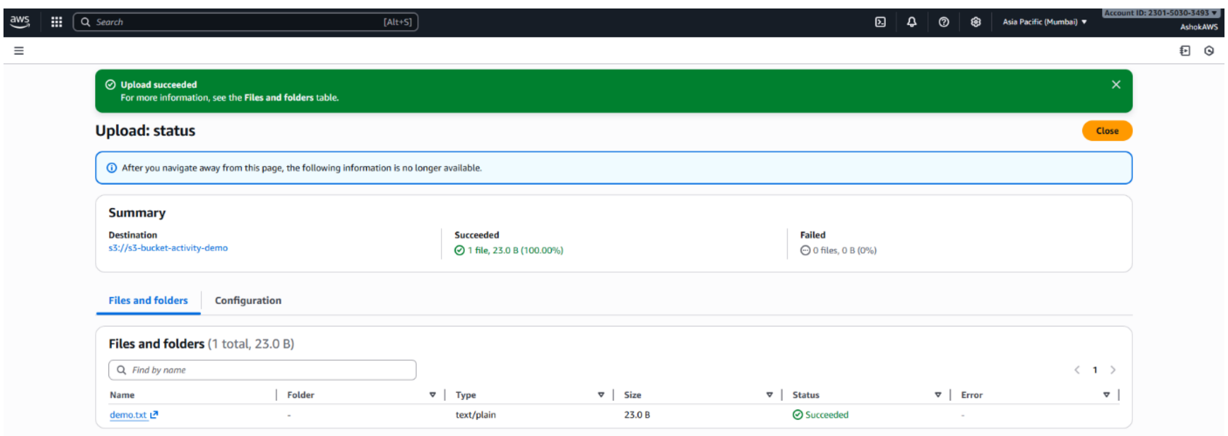
<table>
<tr><td></td><td>
</td></tr>
<tr><td>5</td><td>Configure Bucket Settings for Object Ownership, Block Public Access settings, Bucket versioning and default encryption

</td></tr>
<tr><td>6</td><td>Review the settings and click Create bucket, success message will be displayed</td></tr>
</table>

<table>
<tr><td></td><td>
</td></tr>
<tr><td>7</td><td>Click on the newly created bucket


Click Upload button


Click Upload → Add files (or Add folder).

Select files from your computer
Click Upload
</td></tr>
</table>

	
8	**A success message** will be displayed **after upload/add files.**  Thus, the **S3 bucket is created and configured successfully**

Activity 2 - Design and implement a sample schema on Amazon RDS

2.1 Prerequisite

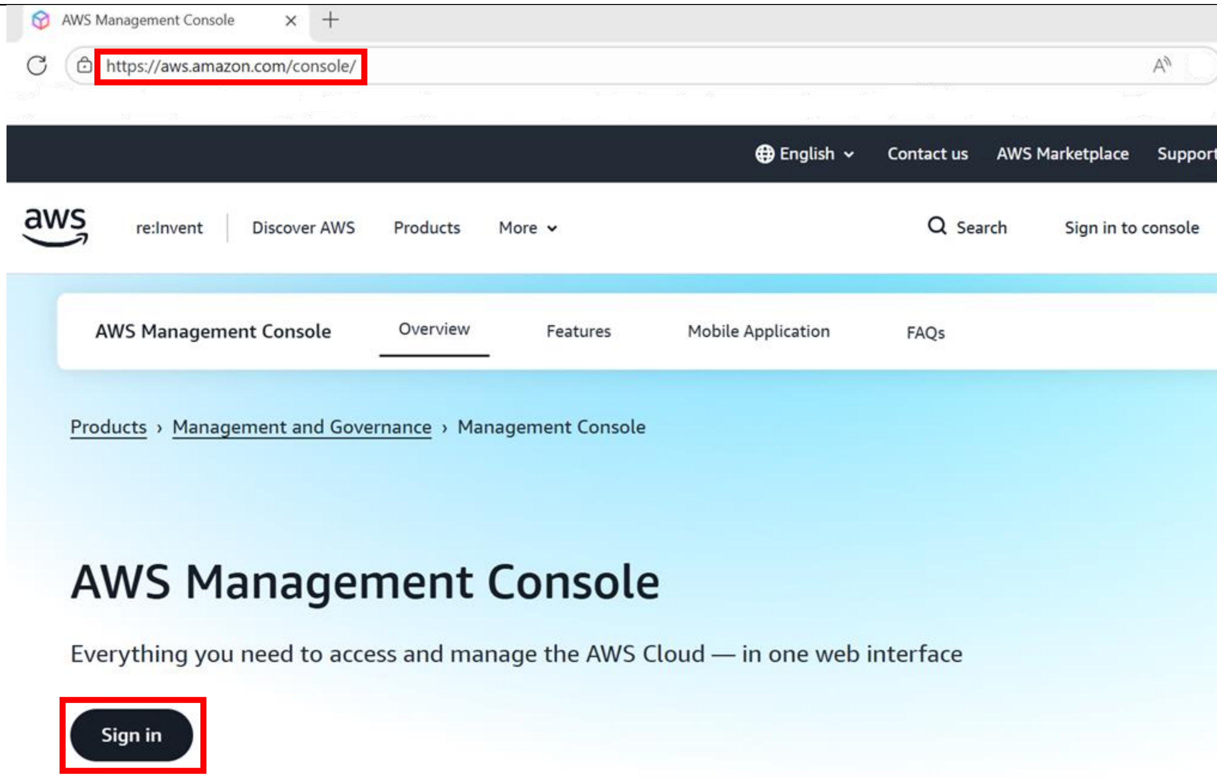
- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

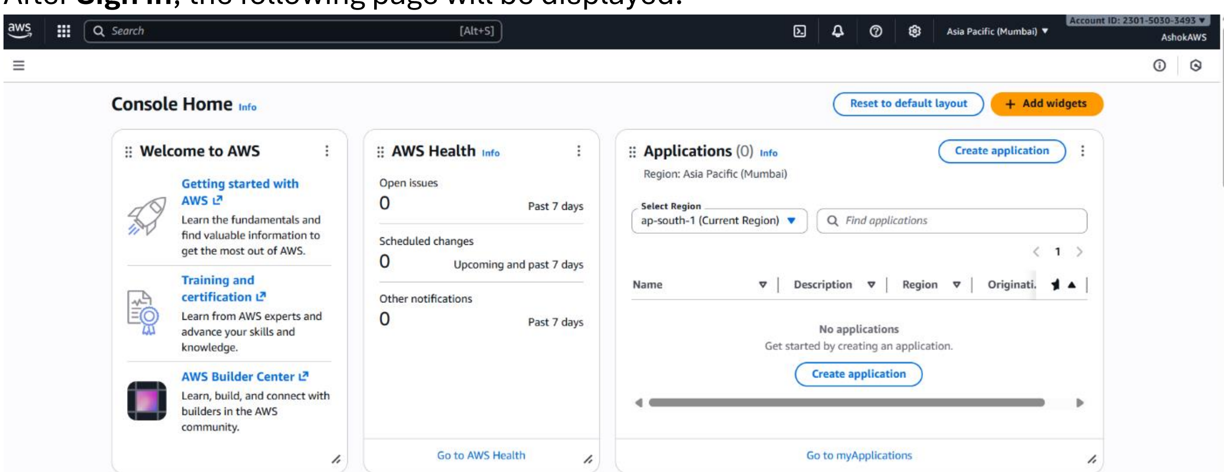
2.2 Scenario

Design and implement a sample schema on Amazon RDS

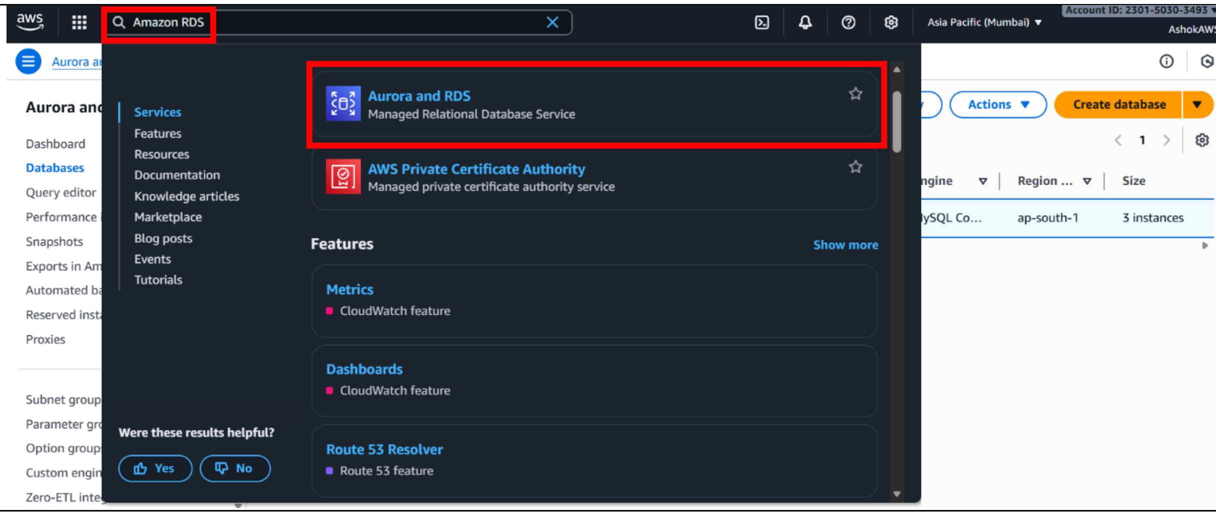
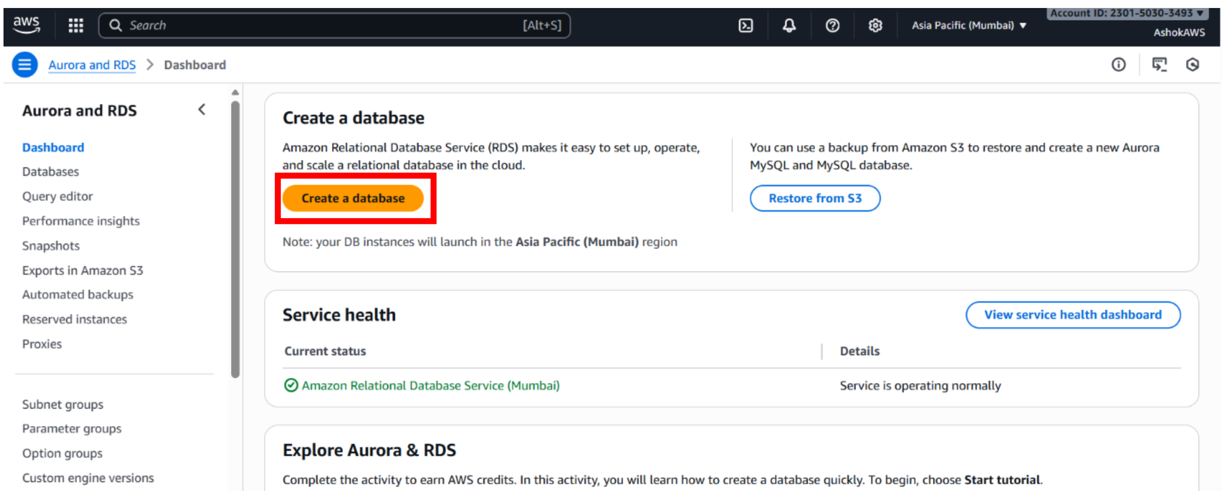
2.3 Steps

1	**Log in to AWS Console** • Visit https://aws.amazon.com/console • Sign in with your credentials

<table>
<tr>
<td></td>
<td>


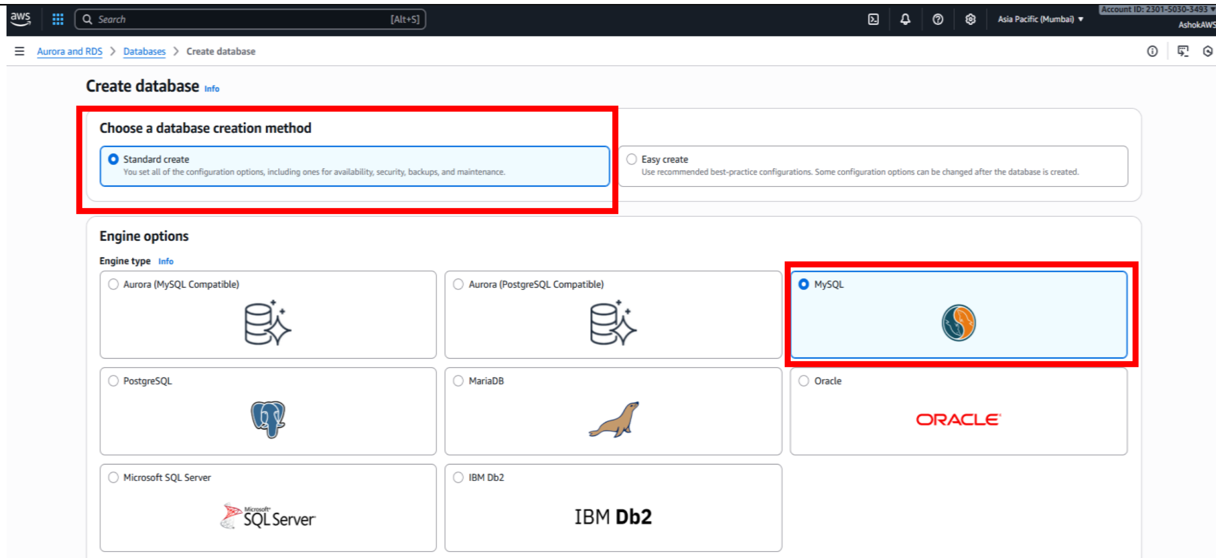
After **Sign in**, the following page will be displayed:


</td>
</tr>
<tr>
<td>2</td>
<td>In the AWS Management Console, search for “Amazon RDS” in the top search bar and click Aurora and RDS</td>
</tr>
</table>

<table>
<tr><td></td><td>
</td></tr>
<tr><td>3</td><td>Click “Create a database” to create the database

</td></tr>
<tr><td>4</td><td>Follow the steps below to create the database:

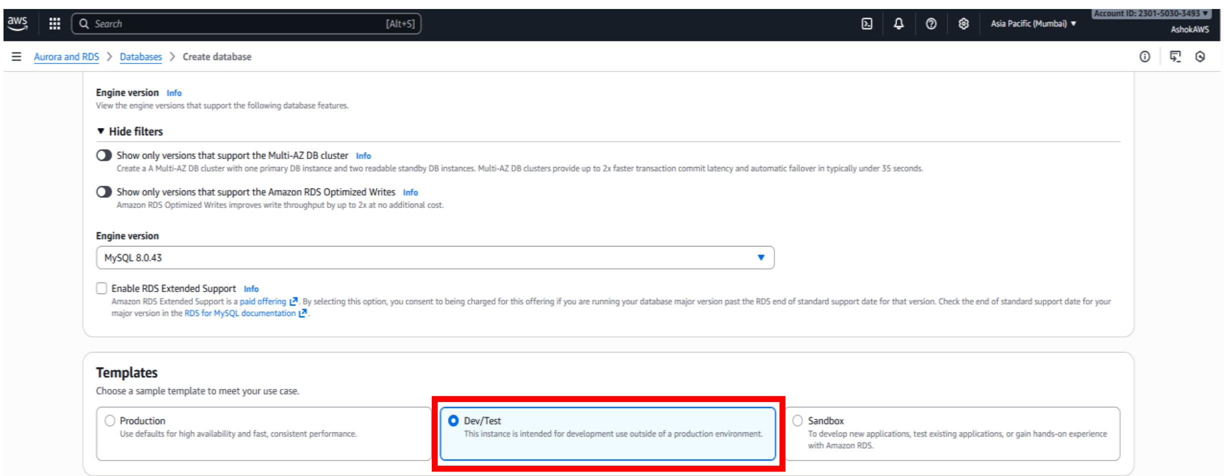
Step 4.1

Under Create database → Click “Standard create” option from the database creation method
Under Engine options → Click the Engine type as “MySQL”
</td></tr>
</table>



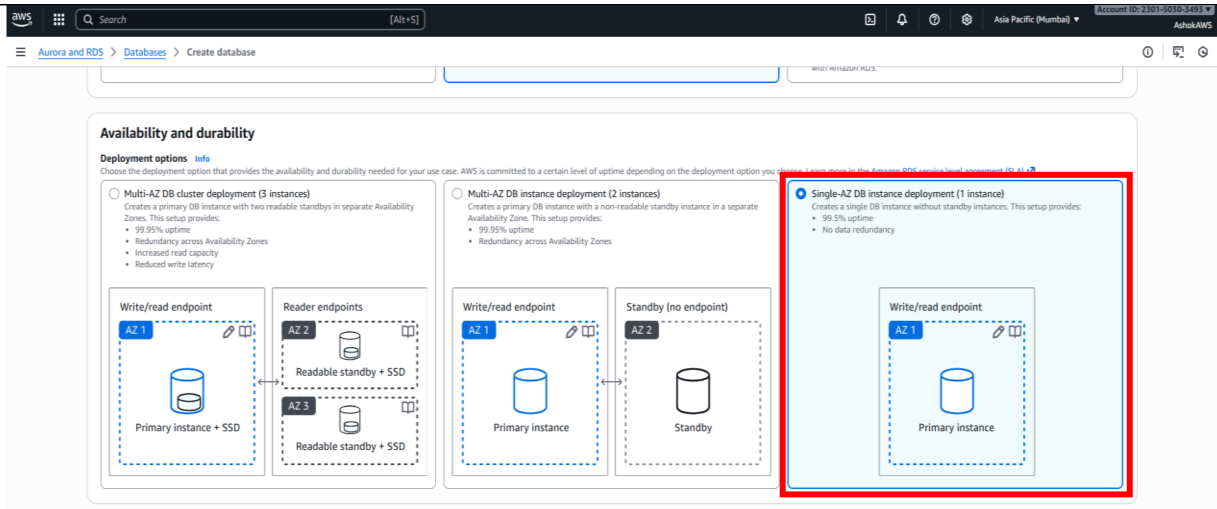
Step 4.2

Under **Templates** → Click **"Dev/Test"** option from the templates



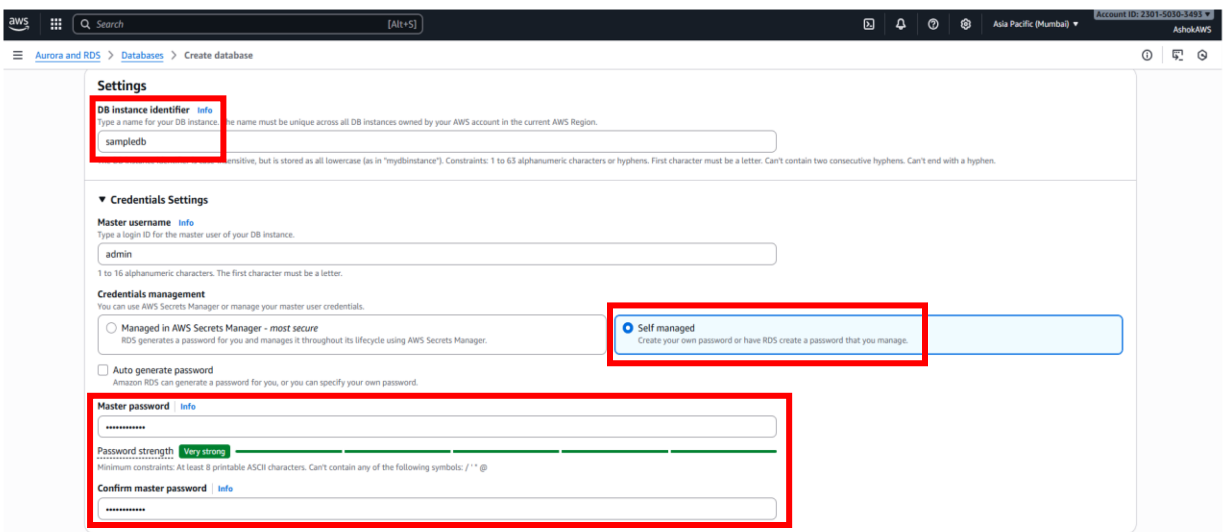
Step 4.3

Under **Availability and durability** (deployment options) → Click **"Single-AZ DB instance deployment (1 instance)"** for the activity



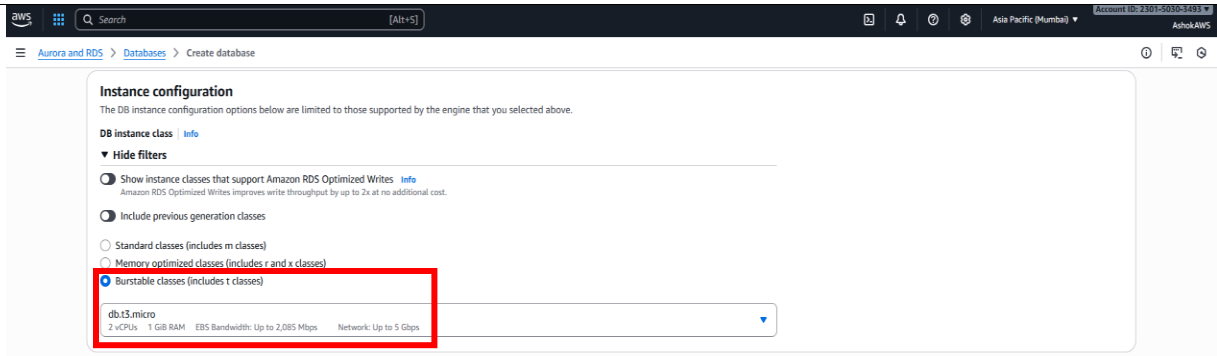
Step 4.4

- Under **Settings → DB instance identifier →** give a name for the DB instance (***e.g., sampledb***)
- Under **Credentials Settings → Credential management →** Click **"Self managed" → Master Password →** give a strong Password and confirm



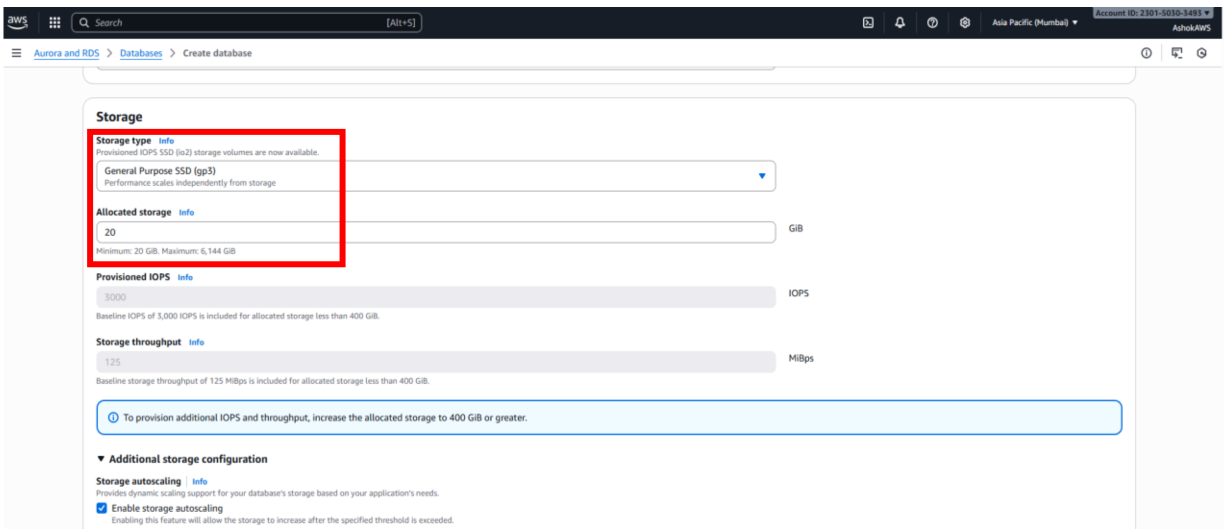
Step 4.5

Under **Instance Configuration →** Click **"Burstable classes (includes t classes)" →** select **db.t3.micro** from the drop-down list



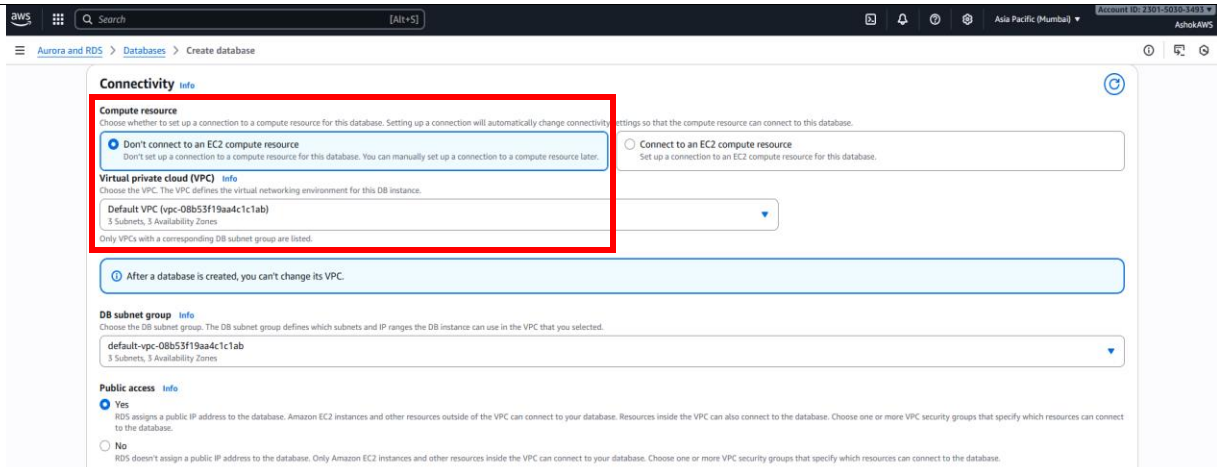
Step 4.6

Under **Storage** → Choose the **Storage type** as **General Purpose SSD (gp3)** and **Allocated storage** as **20**



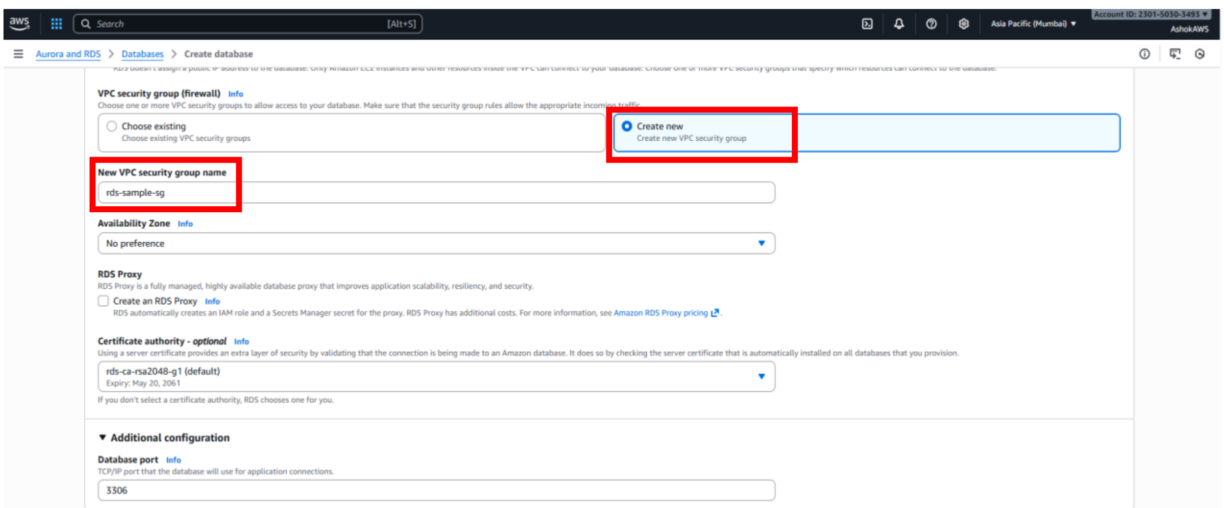
Step 4.7

Under **Connectivity** → Click "**Don't connect to an EC2 compute resource**" and Choose **VPC** as **Default VPC**



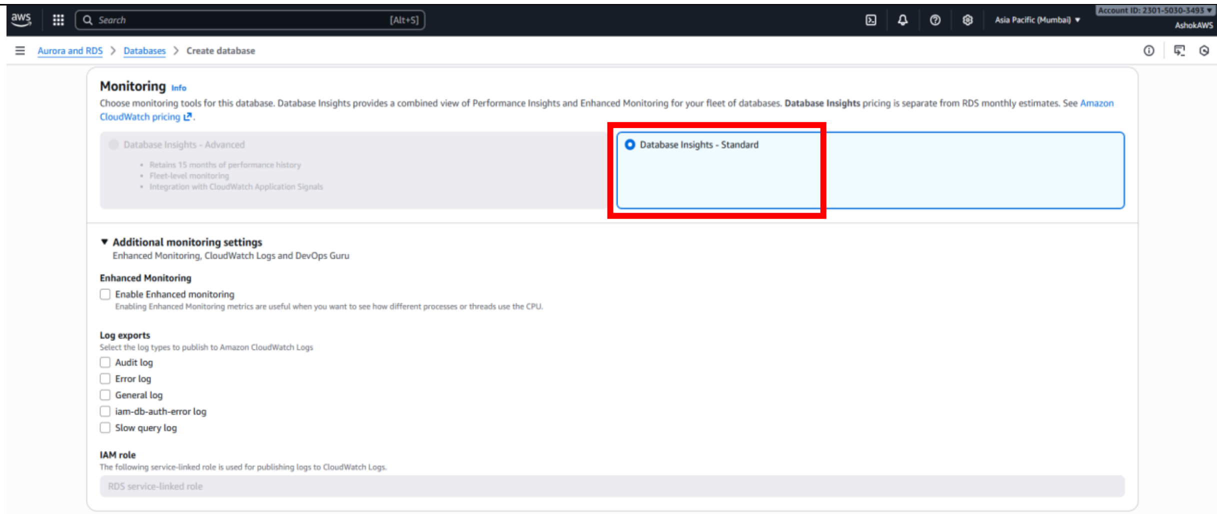
Step 4.8

Under **VPC security group (firewall) →** Click **"Create new"** and give a name for the VPC security group name ***(e.g., rds-sample-sg***)



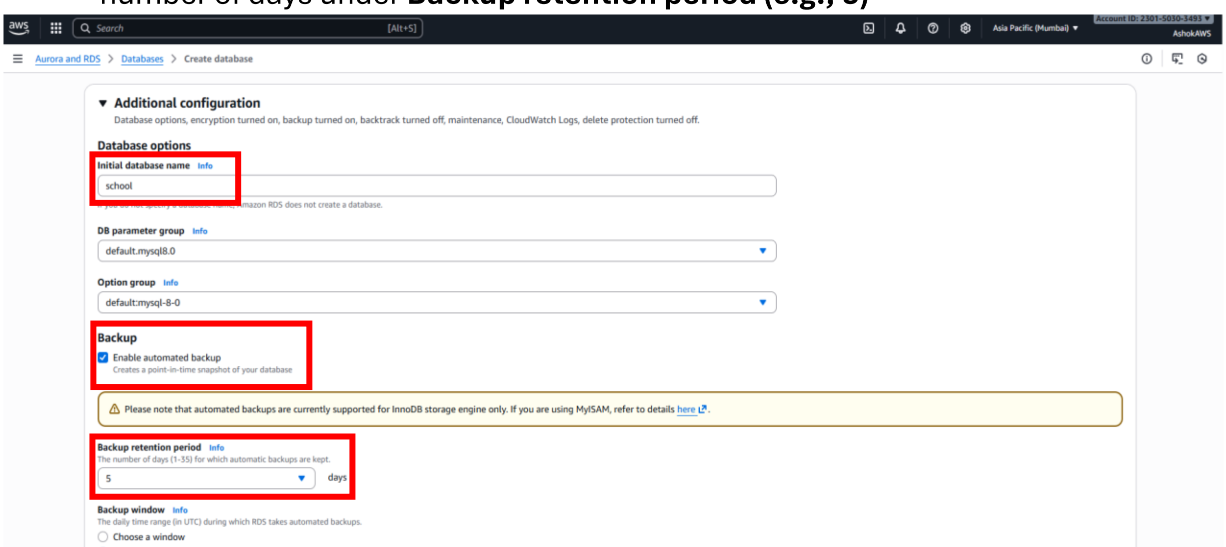
Step 4.9

Under **Monitoring →** Click **"Database Insights – Standard"**



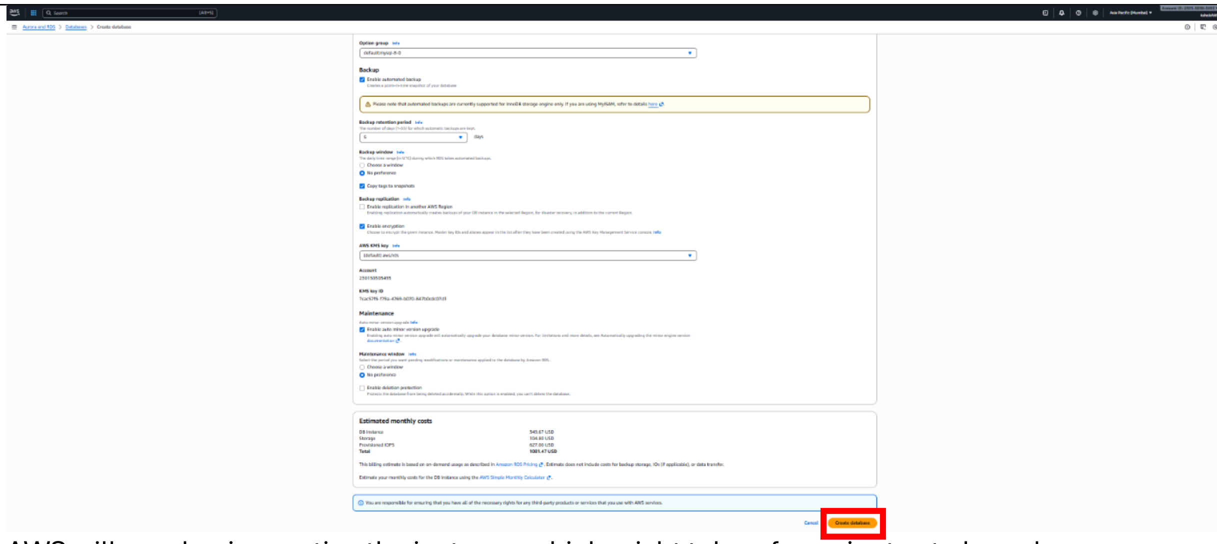
Step 4.10

- Under **Additional configuration → Database options →** give a name for the initial database name ***(e.g., school)***
- Under **Backup → "Check the option: Enable automated backup"** and give number of days under **Backup retention period (*e.g., 5*)**

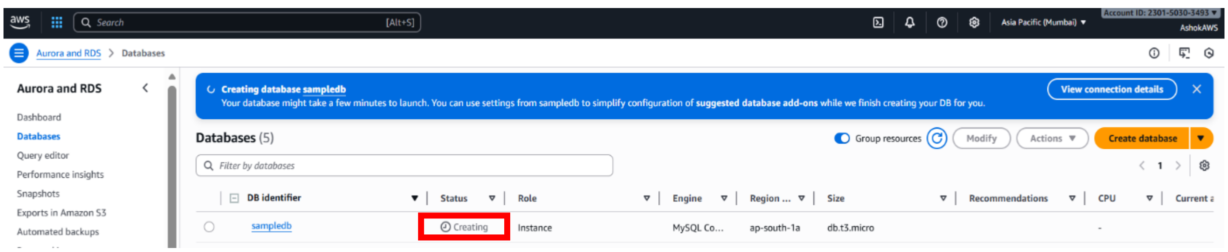


Step 4.11

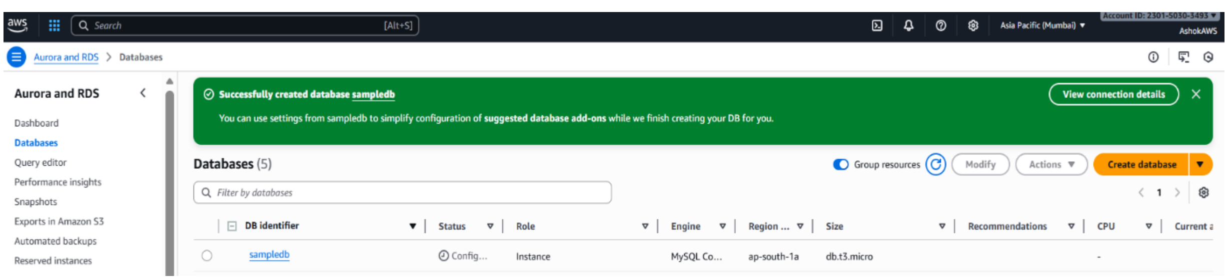
After the above-mentioned steps, click **"Create database" button**



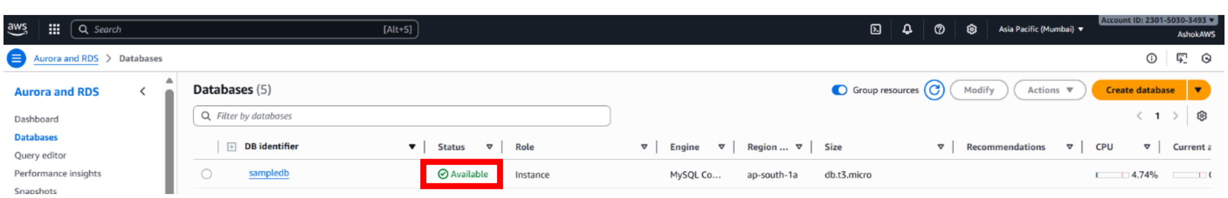
AWS will now begin creating the instance which might take a few minutes to launch.



A success message will be displayed **after the database is created.**



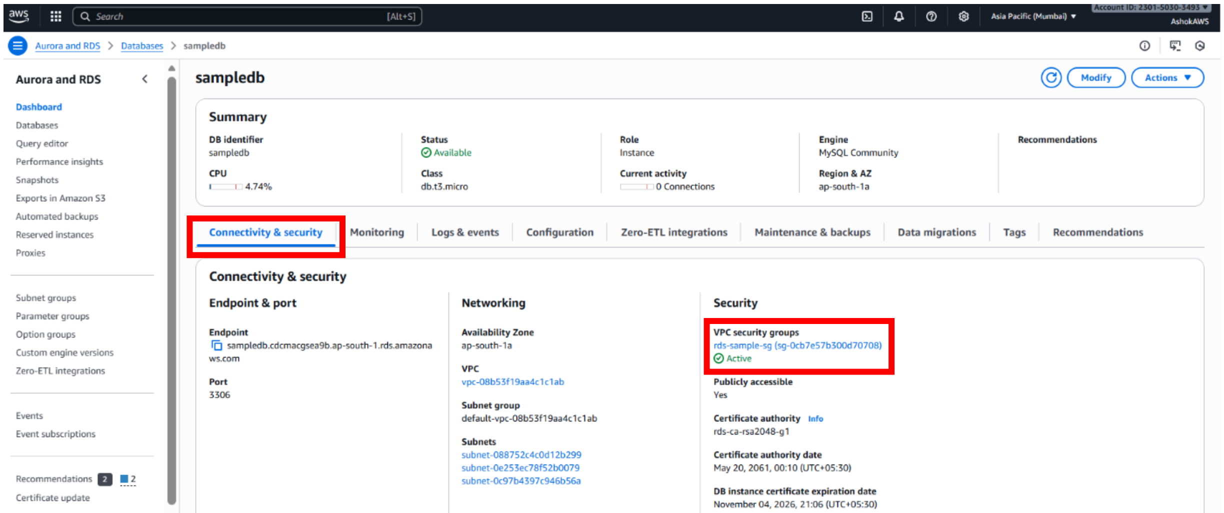
Once the DB status shows **“Available”**:



Follow the steps:

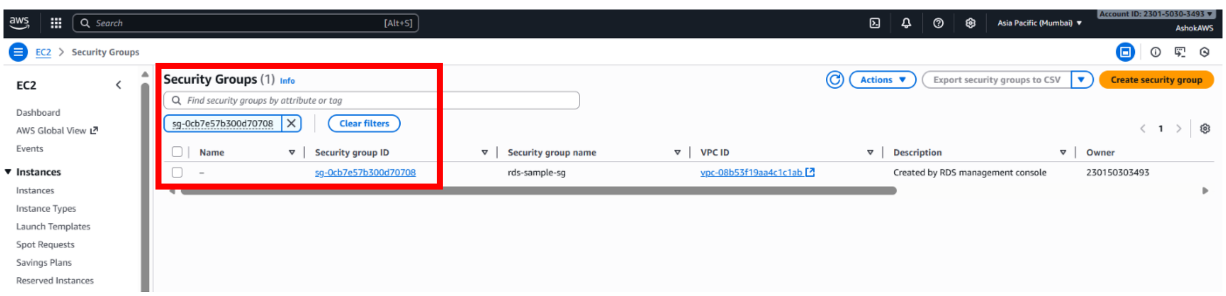
Step 4.12

Go to the **Connectivity & security** tab and find the **VPC security groups** →Click the link (shown).



Step 4.13

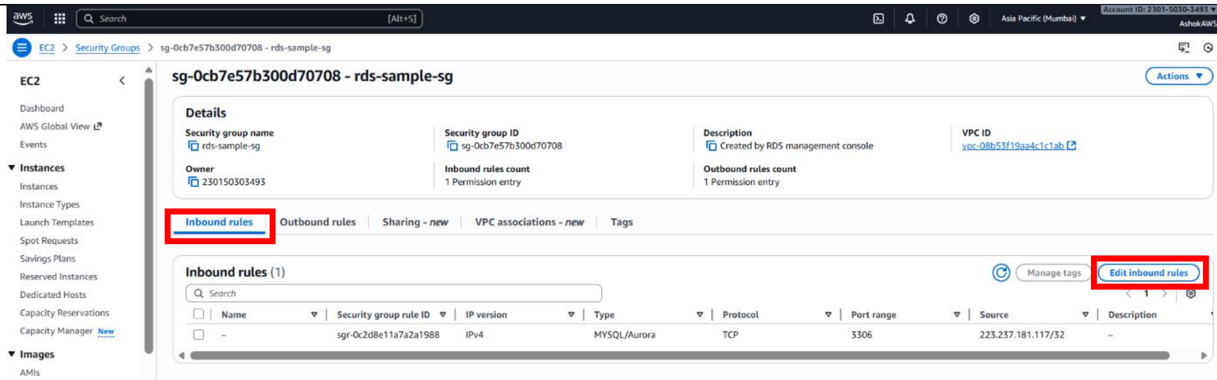
In the **Security Groups** page → Click the ID under **Security group ID**

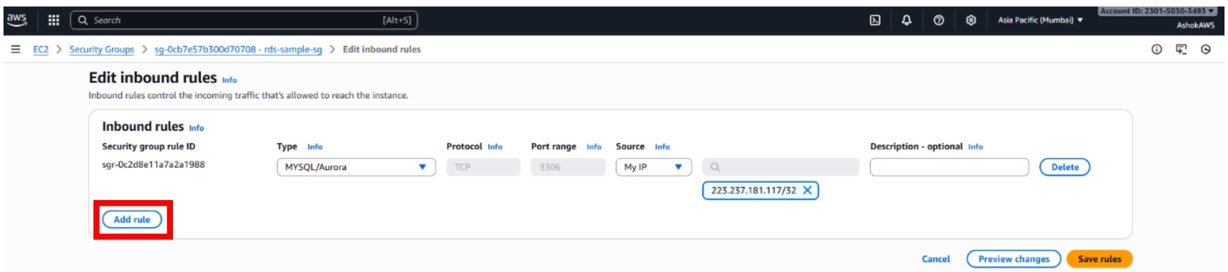


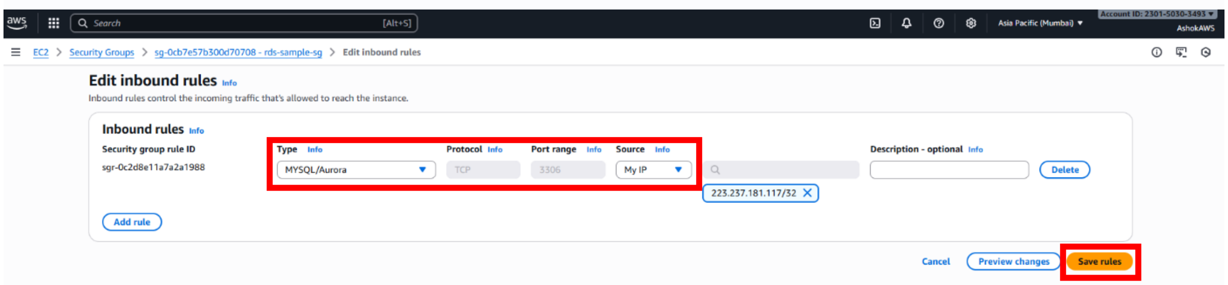
Step 4.14

Under **Inbound rules** tab → Click **"Edit inbound rules"** → Click **"Add rule"** and add the given details:

- **Type:** MySQL/Aurora (or PostgreSQL)
- **Port:** 3306 (for MySQL) or 5432 (for PostgreSQL)
- **Source: My IP**
- Click **"Save rules"**



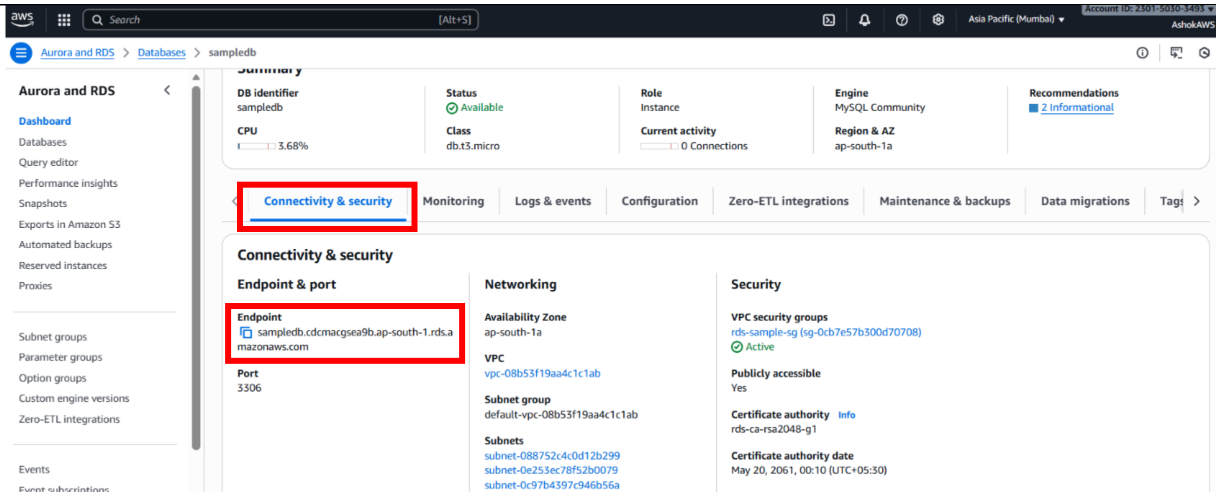




This makes sure **only your computer** can connect to the database.

Step 4.15

After the above-mentioned steps, go back to your RDS instance → **Connectivity & security tab** → Copy the **Endpoint** ***(e.g., sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com)***



You'll use the **Endpoint** to connect like this in **Command-Line Interface**:
mysql -h sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com -u admin -p

Note: In **Step 4.10**, the database name ***"school"*** was created.

Let's create a simple ***"school database"*** with:

- **Students** (student_id, name, email)
- **Courses**(course_id, title, credits)
- **Enrollments** (enrollment_id, student_id, course_id, grade)

5 **Connect to the RDS Database**

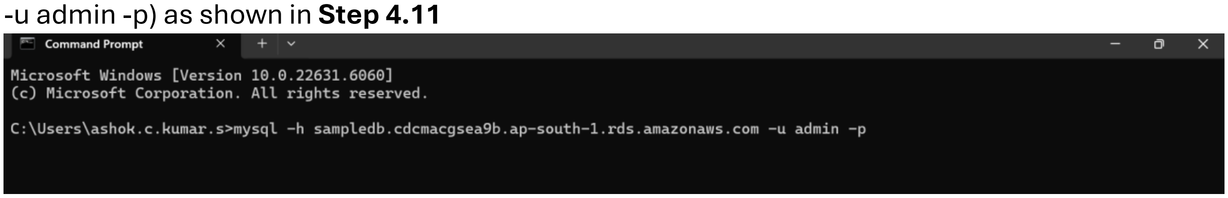
Once your RDS instance is available:

1. Find the **Endpoint** in the RDS console ***(e.g., sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com).***
2. Use a SQL client such as:
   - **MySQL Workbench**
   - **pgAdmin**
   - **DBeaver**
   - Or the terminal (**mysql** or **psql** command-line client).

Here, we took **MySQL CLI** (Command-Line Interface)

Step 1: Search for **"cmd" (Command Prompt)** in Start menu

Step 2: Type the **Endpoint** (mysql -h sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com -u admin -p) as shown in **Step 4.11**



Step 3: It will prompt for a **password;** enter the **Master password** given in **Step 4.4**

```
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ashok.c.kumar.s>mysql -h sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com -u admin -p
Enter password:
```

Step 4: After password is entered, click **Enter** key

```
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ashok.c.kumar.s>mysql -h sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com -u admin -p
Enter password: *************
```

Step 5: mysql command is ready for execution

```
Microsoft Windows [Version 10.0.22631.6060]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ashok.c.kumar.s>mysql -h sampledb.cdcmacgsea9b.ap-south-1.rds.amazonaws.com -u admin -p
Enter password: *************
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 91
Server version: 8.0.43 Source distribution

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

6	**Create the Database and Schema** Once connected, run SQL commands to create your schema.

```
mysql> CREATE DATABASE school;
Query OK, 1 row affected (0.07 sec)

mysql> USE school;
Database changed
mysql>
mysql> CREATE TABLE Students (
    ->     student_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     name VARCHAR(100),
    ->     email VARCHAR(100) UNIQUE
    -> );
Query OK, 0 rows affected (0.12 sec)

mysql>
mysql> CREATE TABLE Courses (
    ->     course_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     title VARCHAR(100),
    ->     credits INT
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql> CREATE TABLE Enrollments (
    ->     enrollment_id INT AUTO_INCREMENT PRIMARY KEY,
    ->     student_id INT,
    ->     course_id INT,
    ->     grade CHAR(1),
    ->     FOREIGN KEY (student_id) REFERENCES Students(student_id),
    ->     FOREIGN KEY (course_id) REFERENCES Courses(course_id)
    -> );
Query OK, 0 rows affected (0.12 sec)
```

Insert Sample Data

```
mysql> INSERT INTO Students (name, email)
    -> VALUES ('Jack', 'jack@example.com'),
    ->        ('Eliza', 'eliza@example.com'),
    ->        ('William', 'william@example.com'),
    ->        ('Maria', 'maria@example.com');
Query OK, 4 rows affected (0.10 sec)
Records: 4  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Courses (title, credits)
    -> VALUES ('Cloud Computing', 4),
    ->        ('Database Systems', 3),
    ->        ('Basic Programming', 4);
Query OK, 3 rows affected (0.08 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO Enrollments (student_id, course_id, grade)
    -> VALUES (1, 1, 'A'),
    ->        (2, 3, 'B'),
    ->        (3, 1, 'B');
Query OK, 3 rows affected (0.07 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

Retrieve the inserted Sample Data:

```
mysql> SELECT * FROM Students;
+------------+---------+---------------------+
| student_id | name    | email               |
+------------+---------+---------------------+
|          1 | Jack    | jack@example.com    |
|          2 | Eliza   | eliza@example.com   |
|          3 | William | william@example.com |
|          4 | Maria   | maria@example.com   |
+------------+---------+---------------------+
4 rows in set (0.06 sec)

mysql> SELECT * FROM Courses;
+-----------+-------------------+---------+
| course_id | title             | credits |
+-----------+-------------------+---------+
|         1 | Cloud Computing   |       4 |
|         2 | Database Systems  |       3 |
|         3 | Basic Programming |       4 |
+-----------+-------------------+---------+
3 rows in set (0.06 sec)

mysql> SELECT * FROM Enrollments;
+---------------+------------+-----------+-------+
| enrollment_id | student_id | course_id | grade |
+---------------+------------+-----------+-------+
|             1 |          1 |         1 | A     |
|             2 |          2 |         3 | B     |
|             3 |          3 |         1 | B     |
+---------------+------------+-----------+-------+
3 rows in set (0.07 sec)
```

Test Your Schema

```
mysql> SELECT s.name, c.title, e.grade
    -> FROM Enrollments e
    -> JOIN Students s ON e.student_id = s.student_id
    -> JOIN Courses c ON e.course_id = c.course_id;
+---------+-------------------+-------+
| name    | title             | grade |
+---------+-------------------+-------+
| Jack    | Cloud Computing   | A     |
| Eliza   | Basic Programming | B     |
| William | Cloud Computing   | B     |
+---------+-------------------+-------+
3 rows in set (0.09 sec)
```

Thus, the **sample schema is designed and implemented on Amazon RDS successfully**

Activity 3 – Create schema on DynamoDB

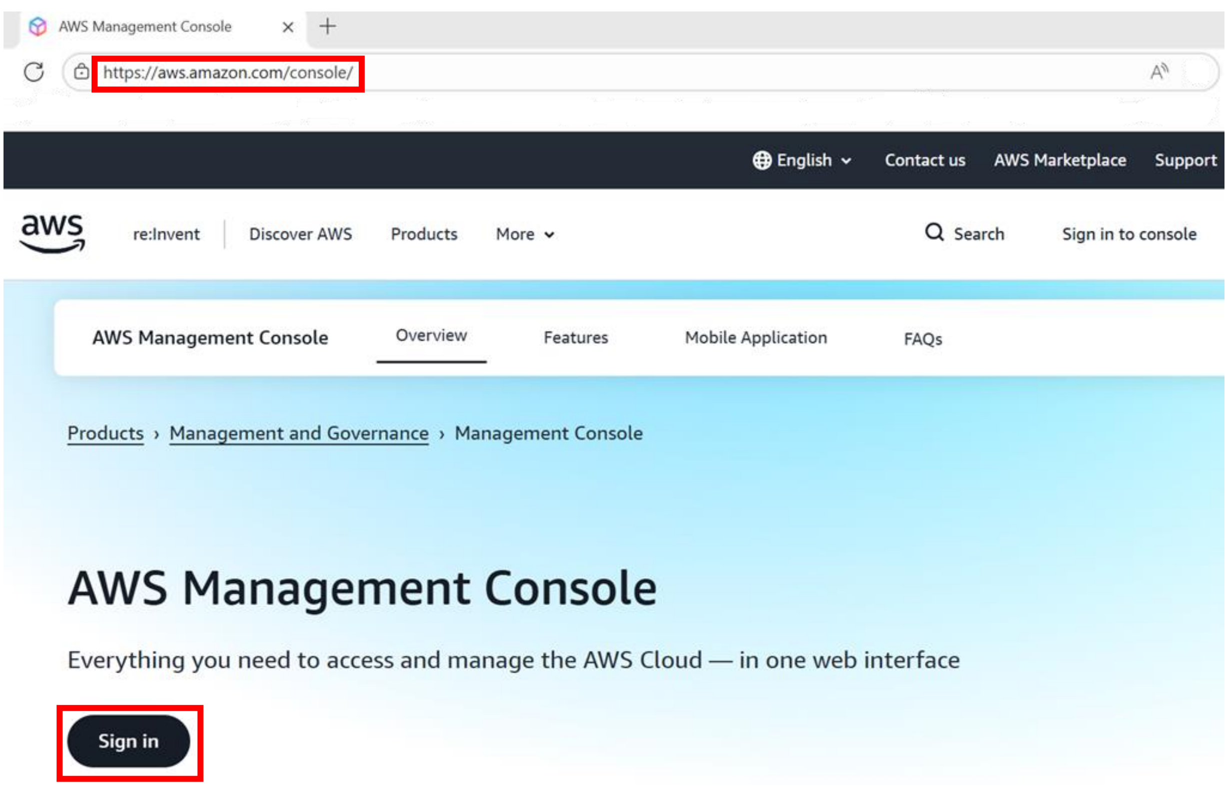
3.1 Prerequisite

- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

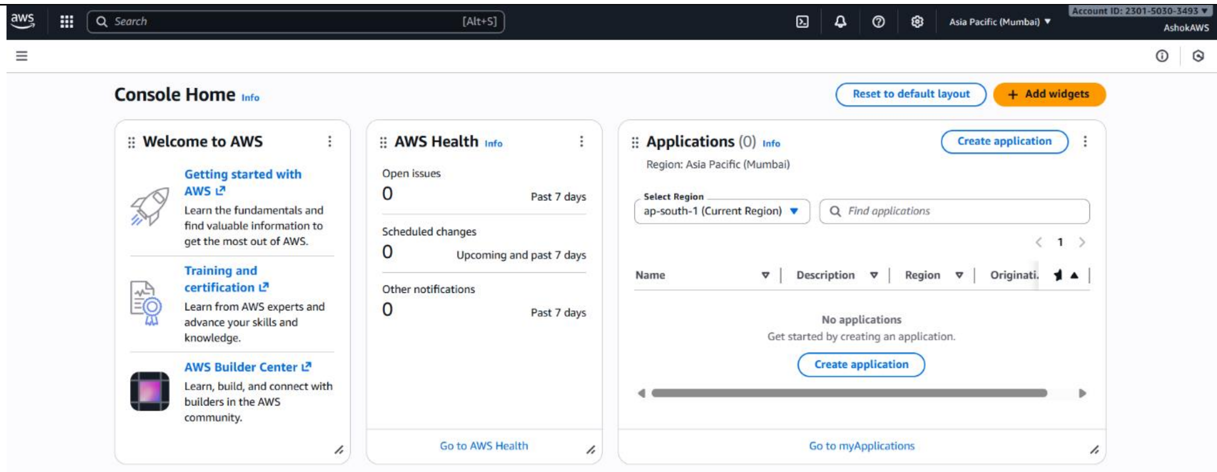
3.2 Scenario

Create schema on DynamoDB, storing and retrieving data

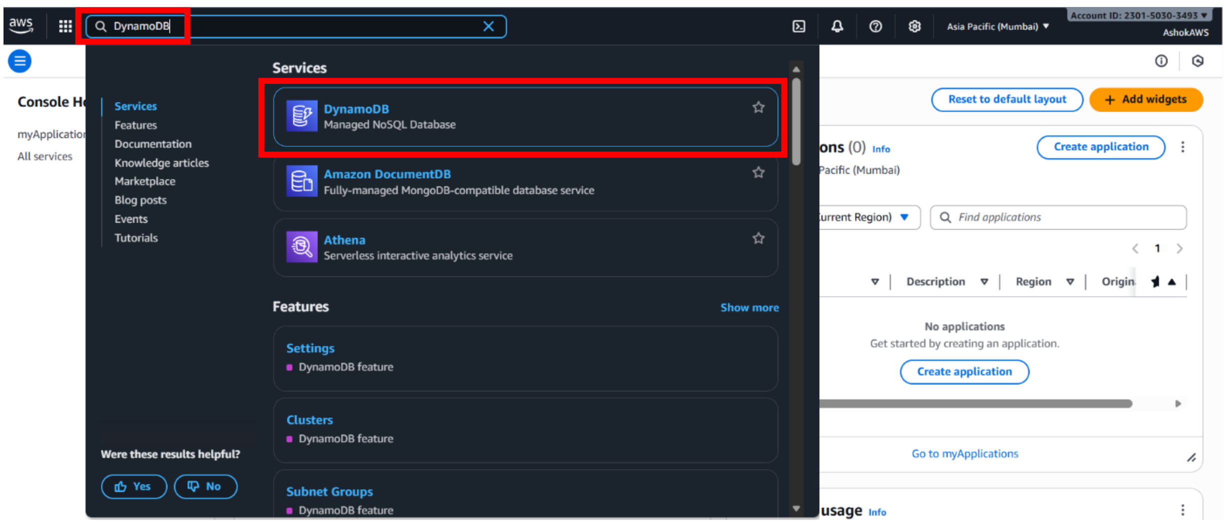
3.3 Steps

<table>
<tr><td>1</td><td>Log in to AWS Console
Visit https://aws.amazon.com/console
Sign in with your credentials


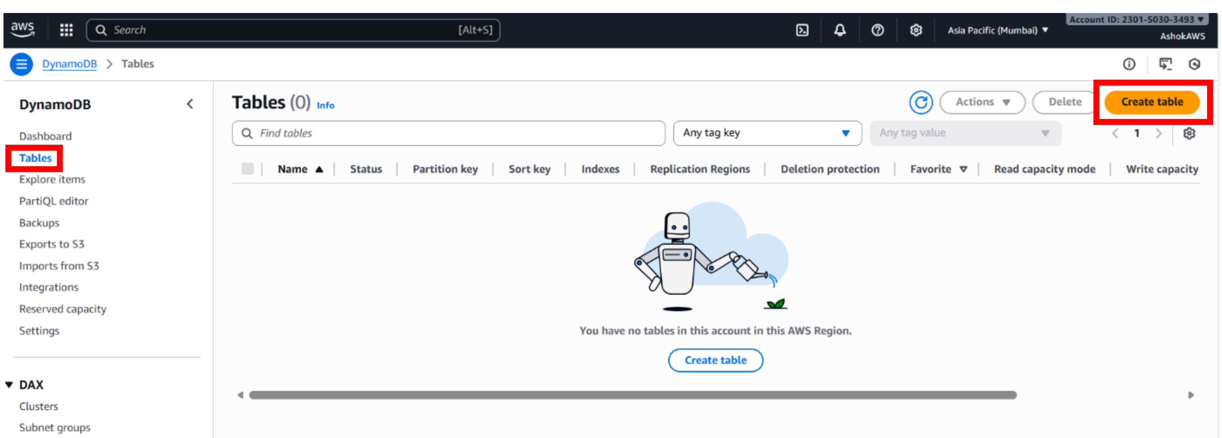
After Sign in, the following page will be displayed:</td></tr>
</table>



2 In the **AWS Management Console**, search for **"DynamoDB"** in the top search bar and click **DynamoDB**



3 In the left-navigation pane, click **Tables** and click the **"Create table" button**

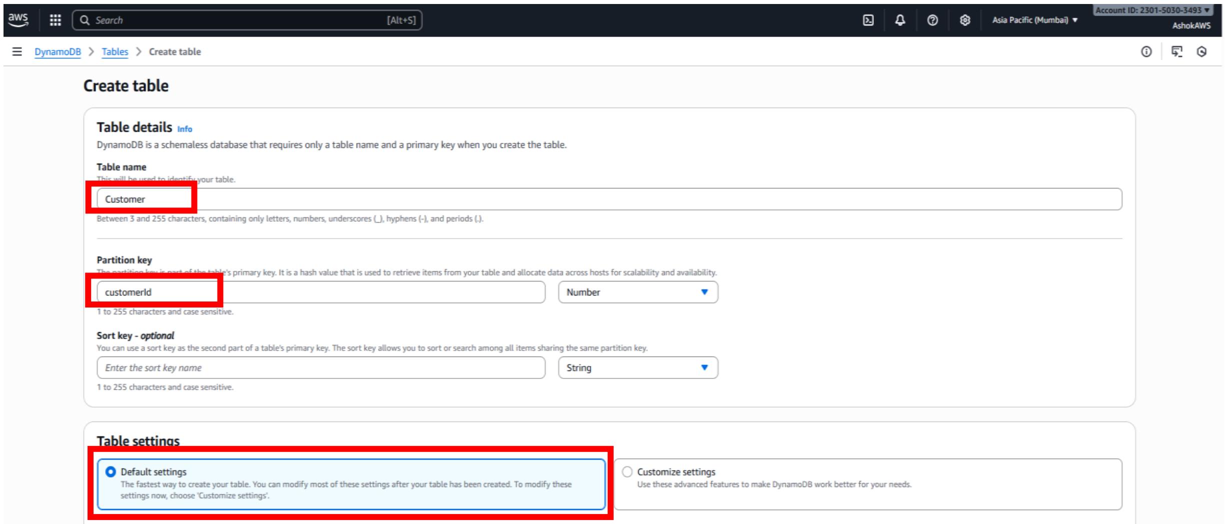


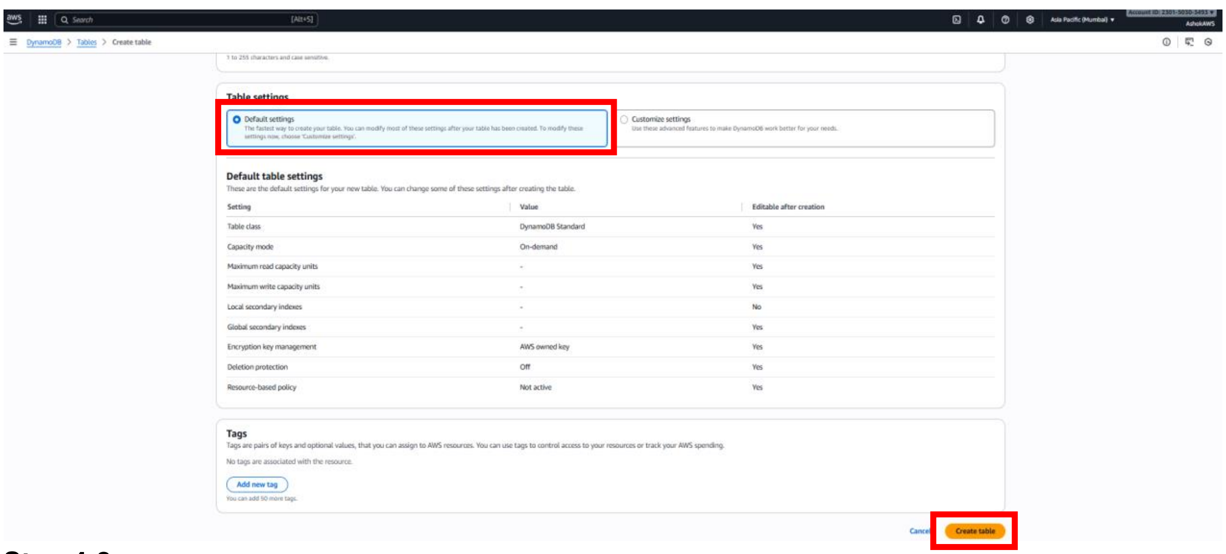
4

On clicking the **"Create table" button**, Enter **Table details – Table name** and **Partition key** and choose **Default settings** under Table settings

Step 4.1

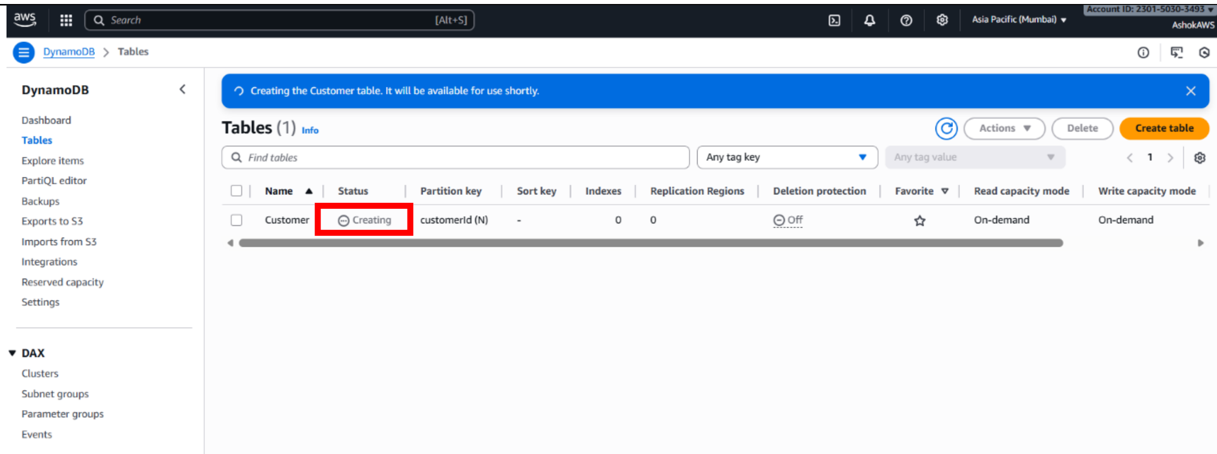
- **Table name**: Enter a meaningful name
- **Partition key**: A mandatory attribute that uniquely identifies items
- Leave defaults and click **Create table**



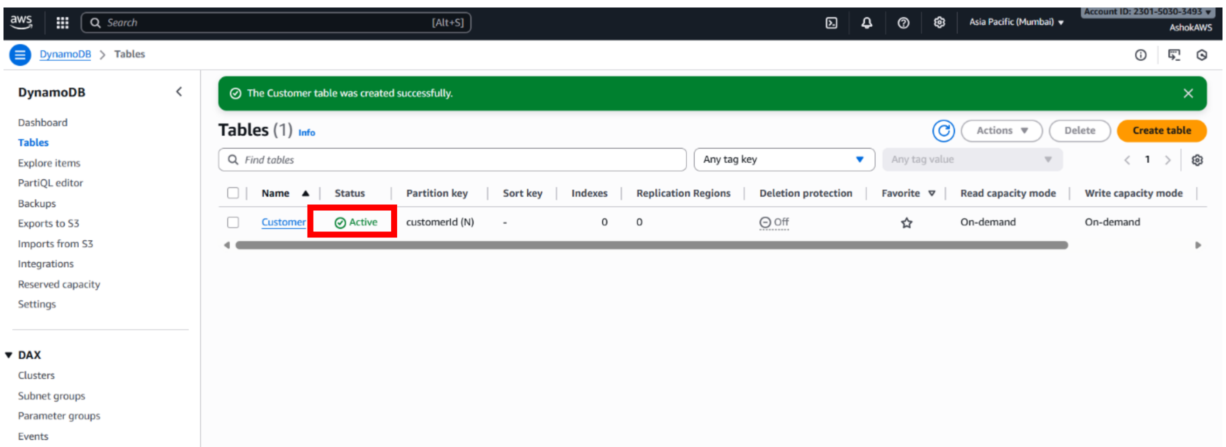


Step 4.2

- After clicking **Create table** button, the table will show up in the **"Tables"** list
- Initially the status will be "**Creating"**

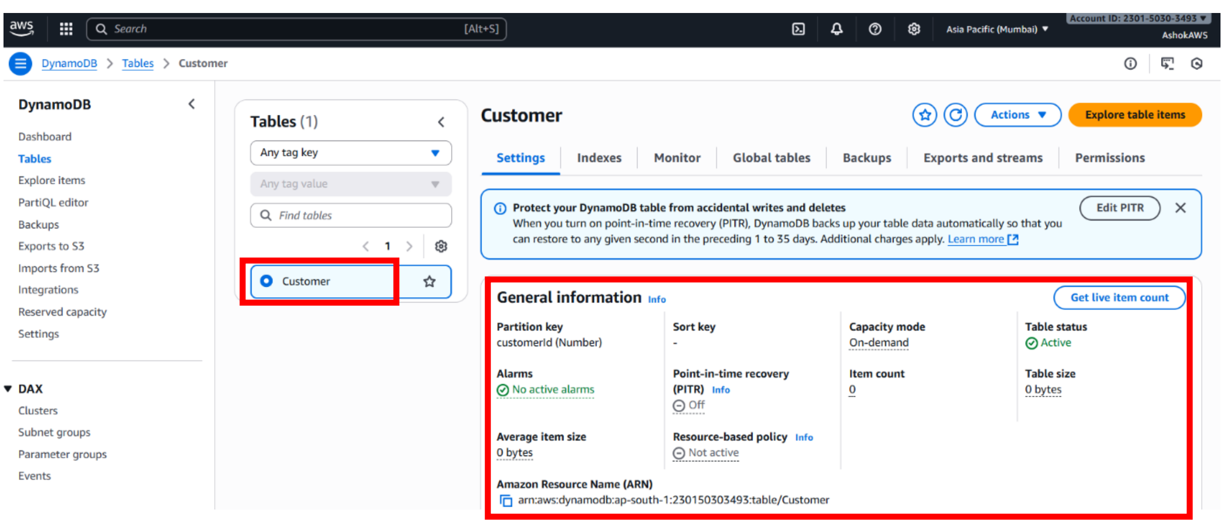


Later, it becomes **"Active"** once ready.



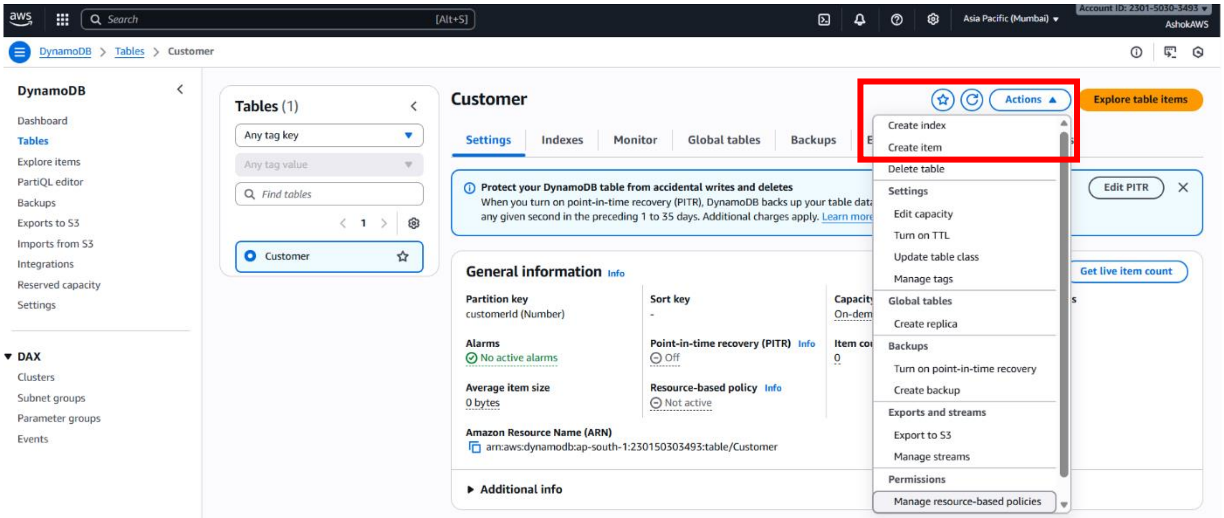
Step 4.3

Once the status is **Active**, click on the table name to view the below details:

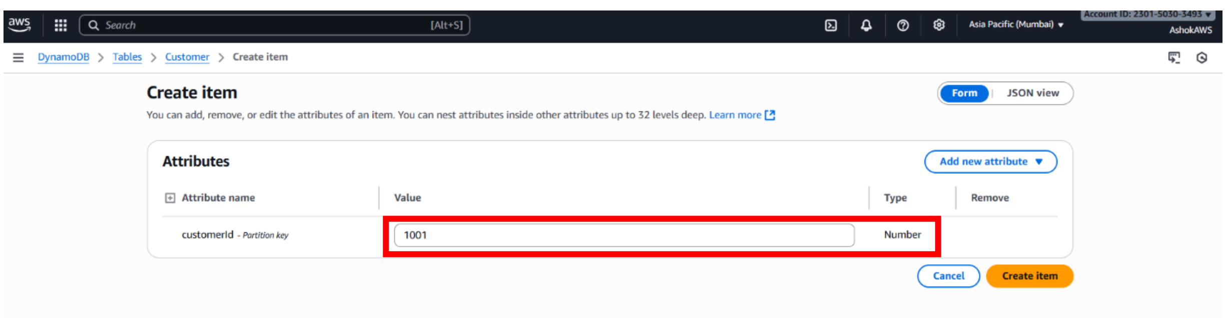


Step 4.4 Store items

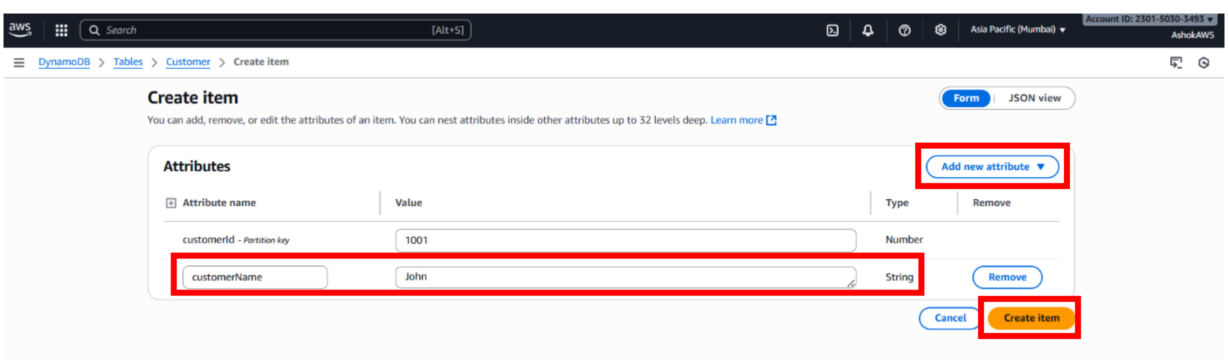
Click **Actions → Create item**



Under **Create item**, add the Value (based on the Type ***e.g., Number, String, etc.,***) for the given Attribute name

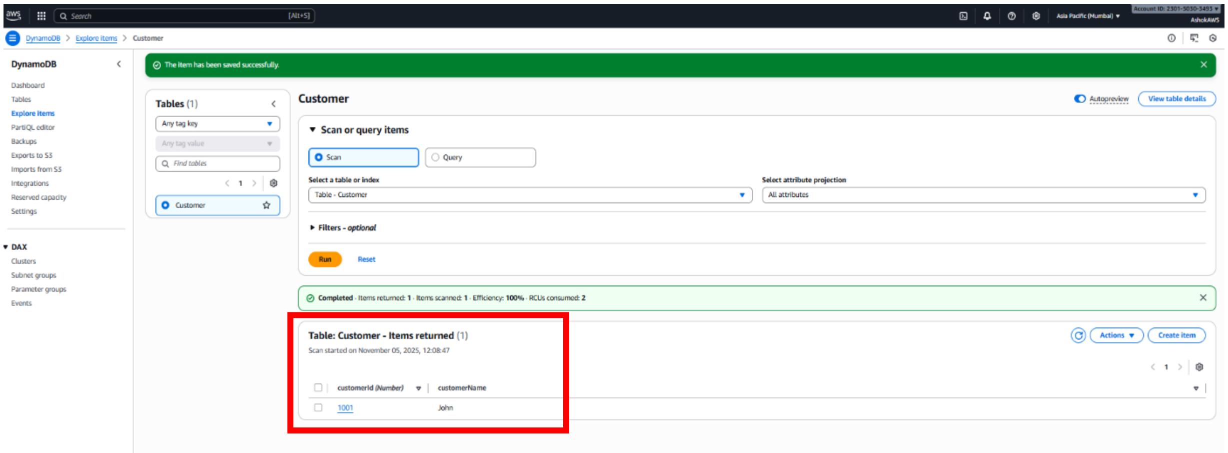


If required, add a new attribute by specifying its Name and Value, and select the Type from the **"Add new attribute"** drop-down list.



Step 4.5 Retrieve items

- After adding the details, click the **“Create Item” button**
- A message will indicate whether the item was saved successfully
- The items will be displayed in the console



Thus, the **schema is created on DynamoDB**, with **data stored and retrieved successfully.**

Activity 4 – Create a VPC with public and private subnets

4.1 Prerequisite

- An **AWS account** (free-tier eligible)
- Access to the **AWS Management Console**
- Basic understanding of AWS services

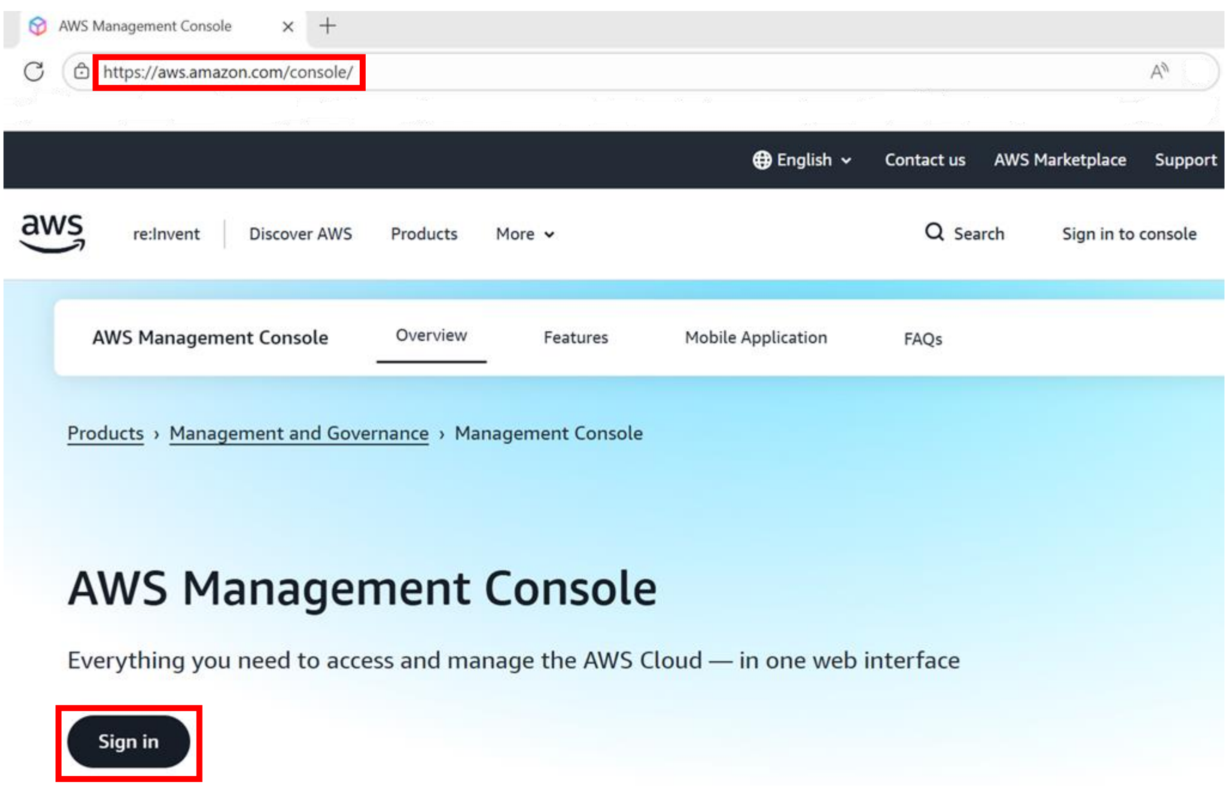
4.2 Scenario

Create a VPC with public and private subnets

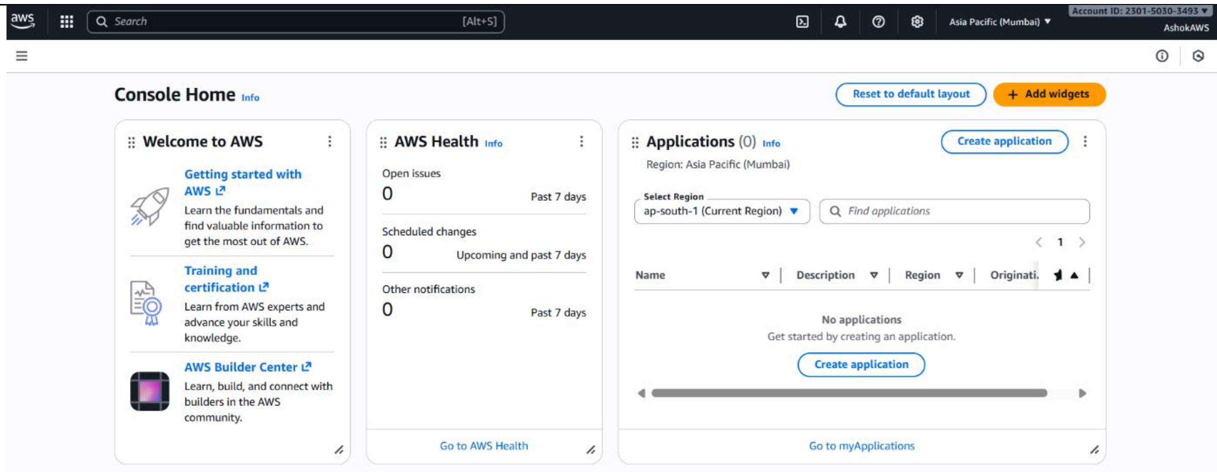
4.3 Steps

<table>
<tr>
<td>1</td>
<td>
Log in to AWS Console

Visit https://aws.amazon.com/console
Sign in with your credentials



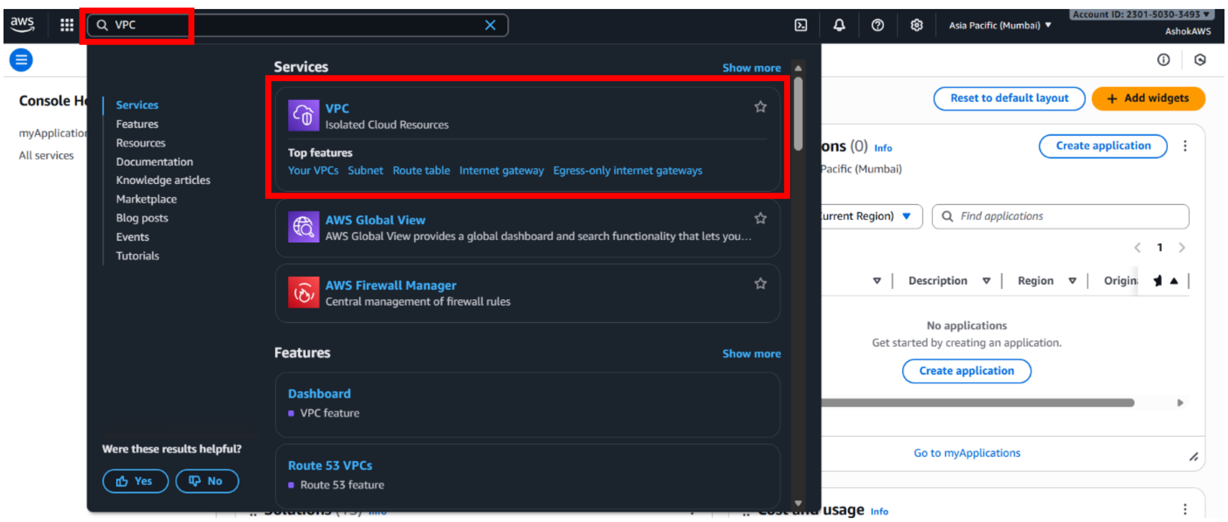
After Sign in, the following page will be displayed:
</td>
</tr>
</table>



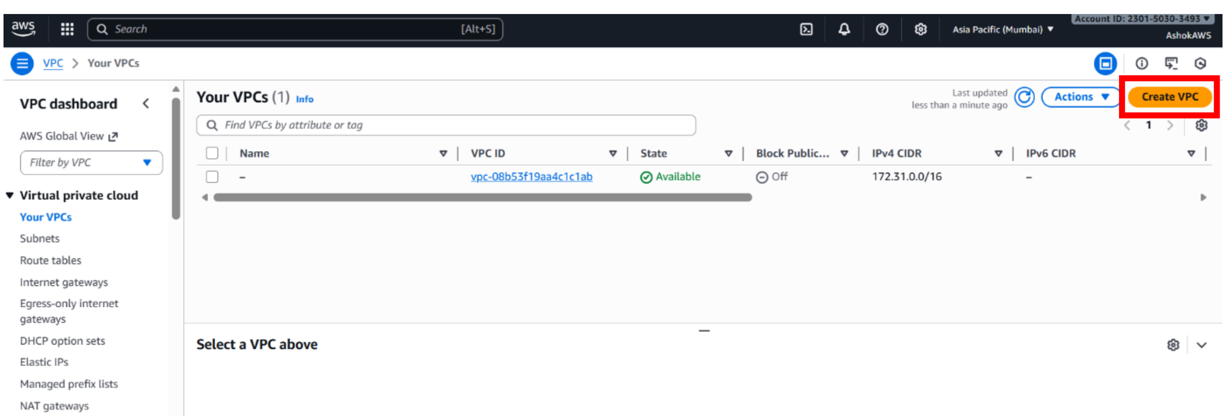
2

Create the VPC

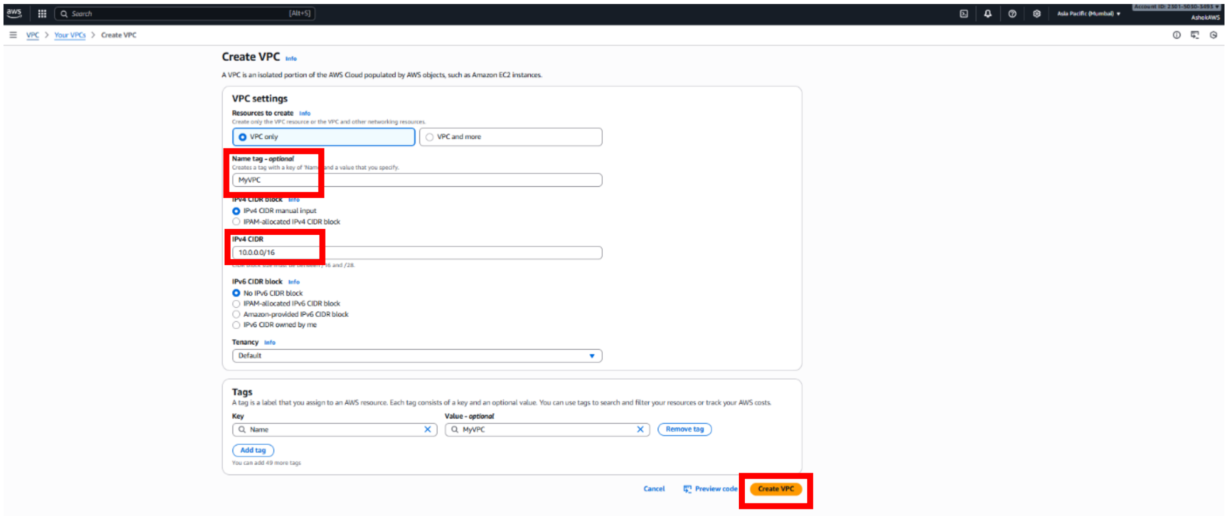
Step 2.1 In the **AWS Management Console**, search for **"VPC"** in the top search bar and click **"Your VPCs"**



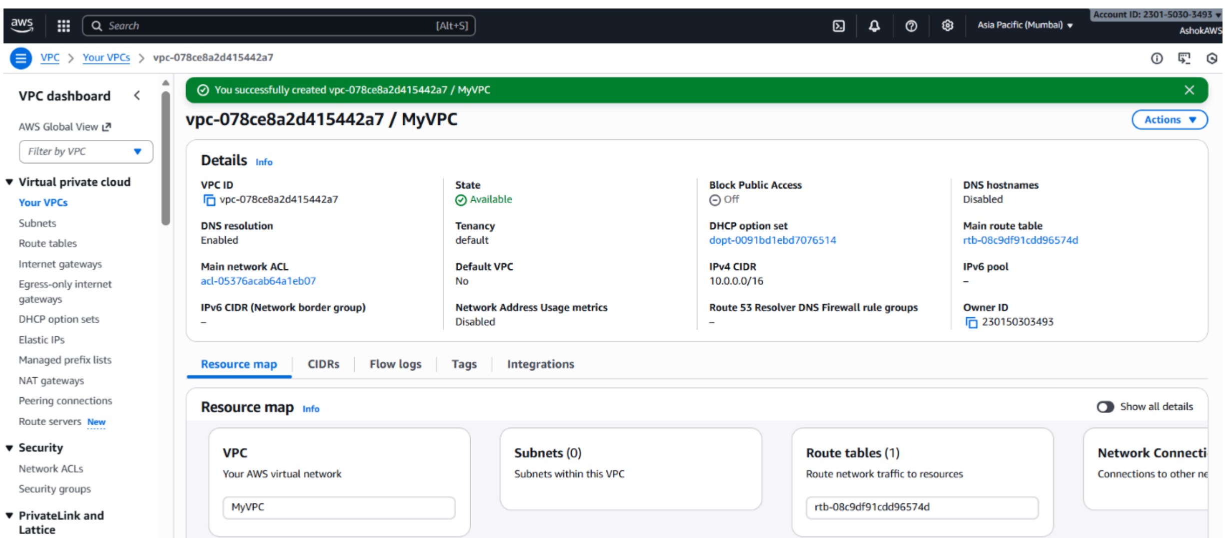
Step 2.2 Click **"Create VPC" button**



Step 2.3 Add the details shown below for the Name tag and IPv4 CIDR. Leave all other settings as default, then click the **"Create VPC" button**



A success message will be displayed **after creating the VPC.**

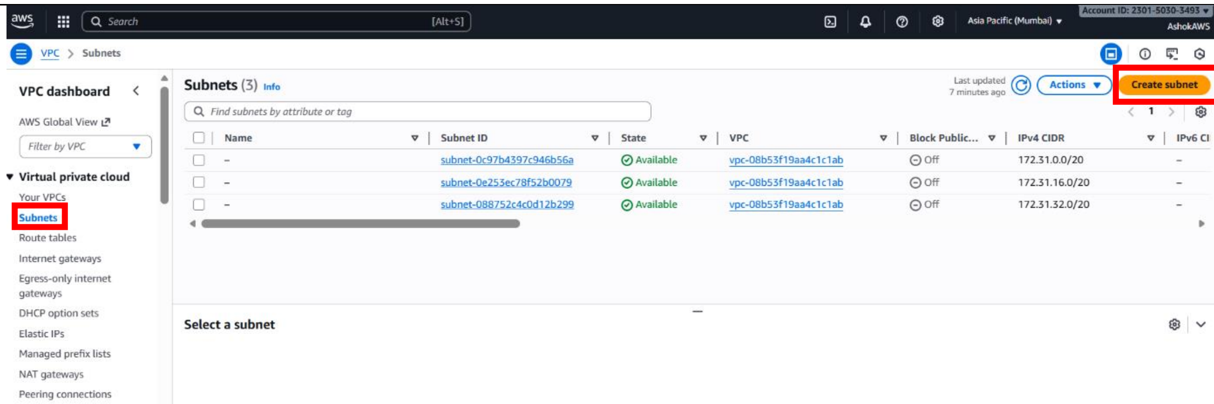


3

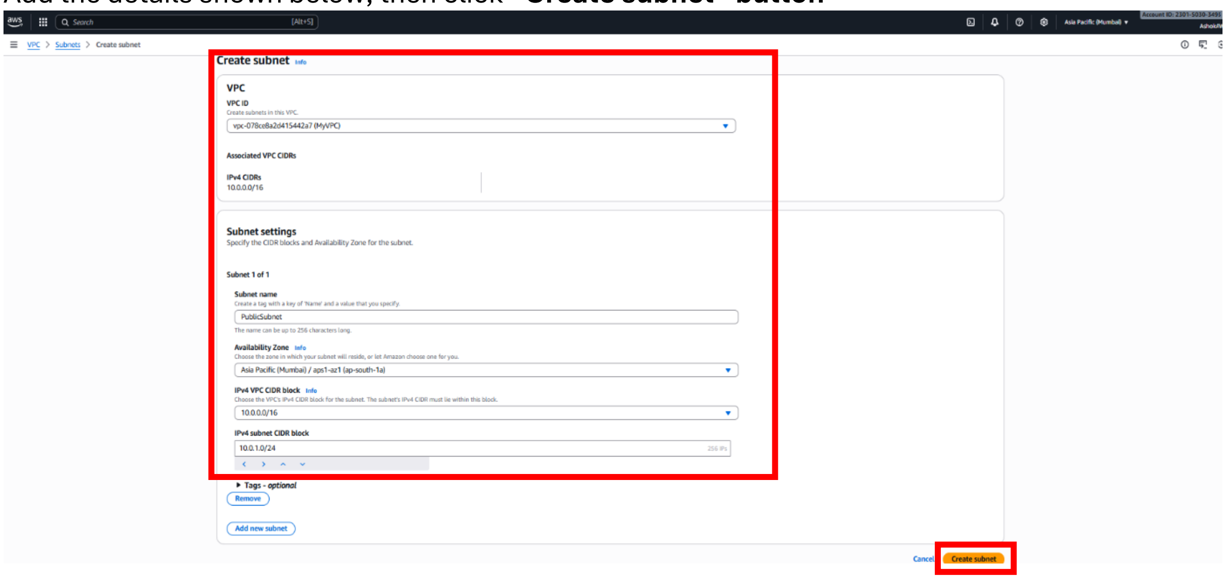
Create Subnets - Create **at least one public and one private subnet.**

Step 3.1 Create Public Subnet

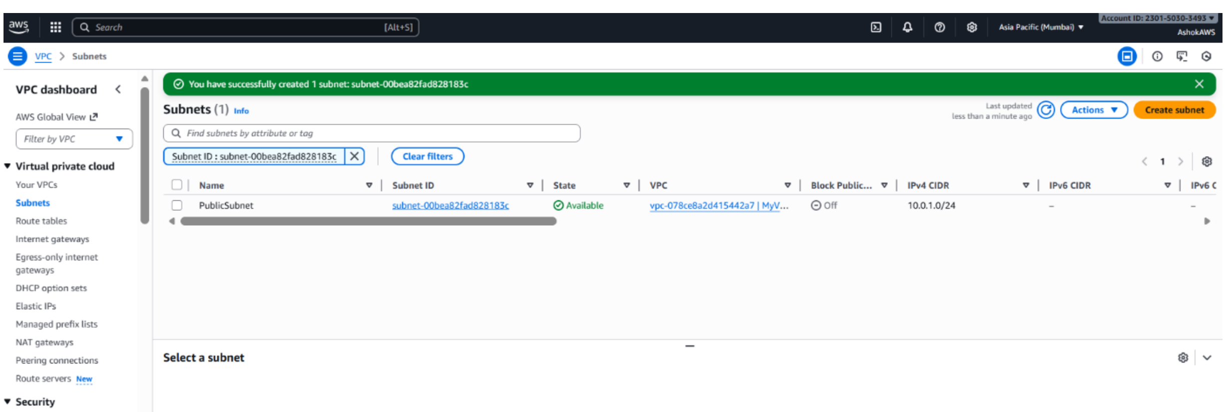
Click **VPC → Subnets → Create Subnet**



Add the details shown below, then click **“Create subnet” button**

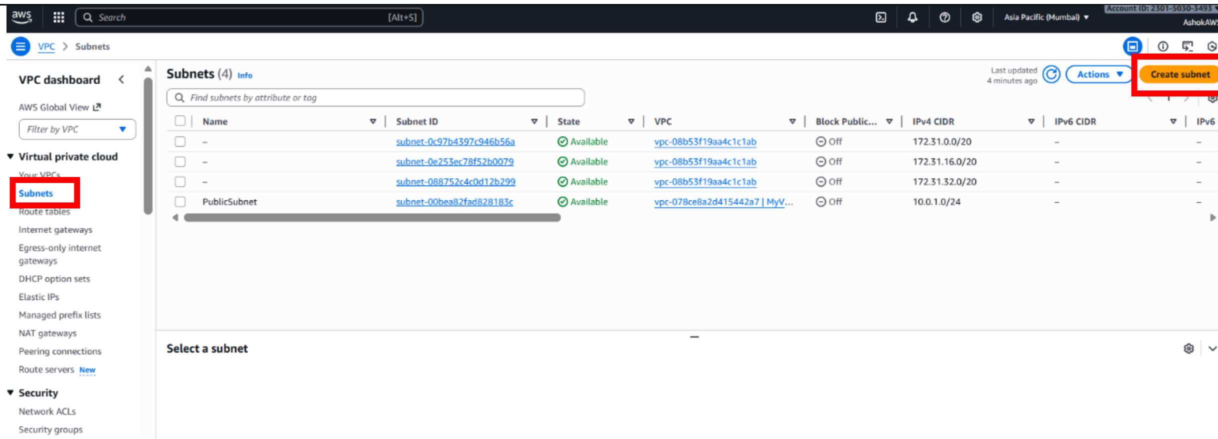


A success message will be displayed **after creating subnet (PublicSubnet).**

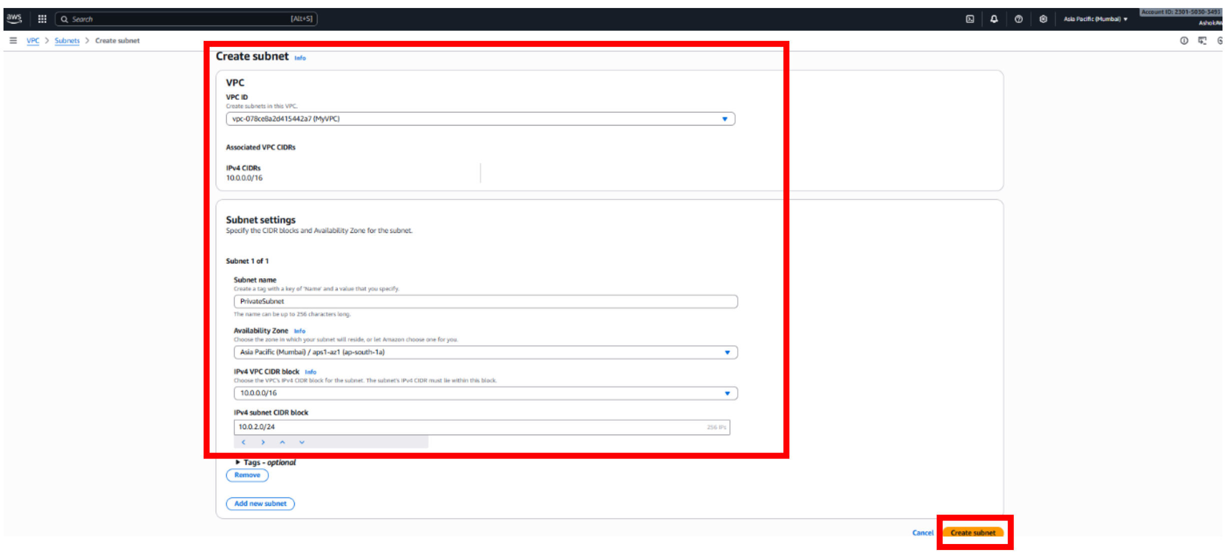


Step 3.2 Create Private Subnet

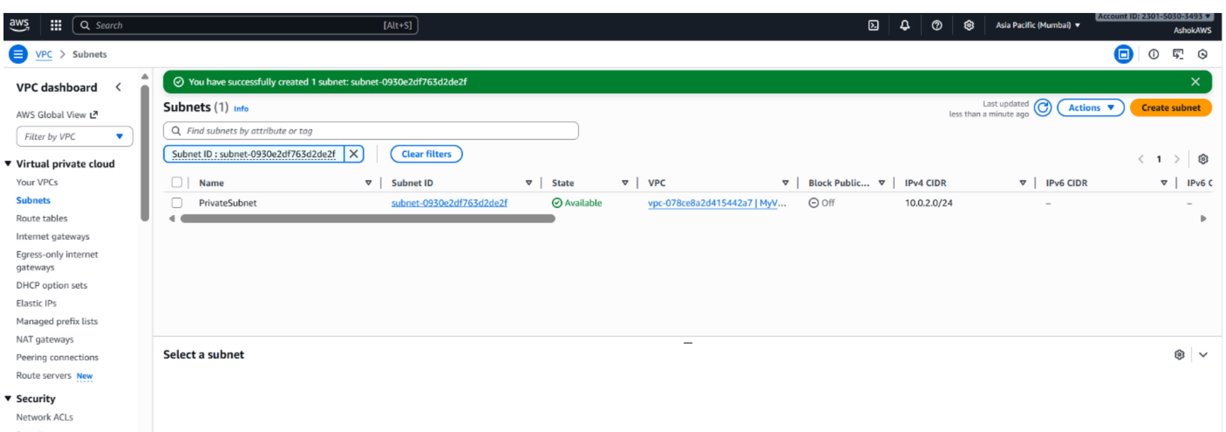
Click **Subnets** → **Create Subnet**



Add the details shown below, then click **"Create subnet" button**

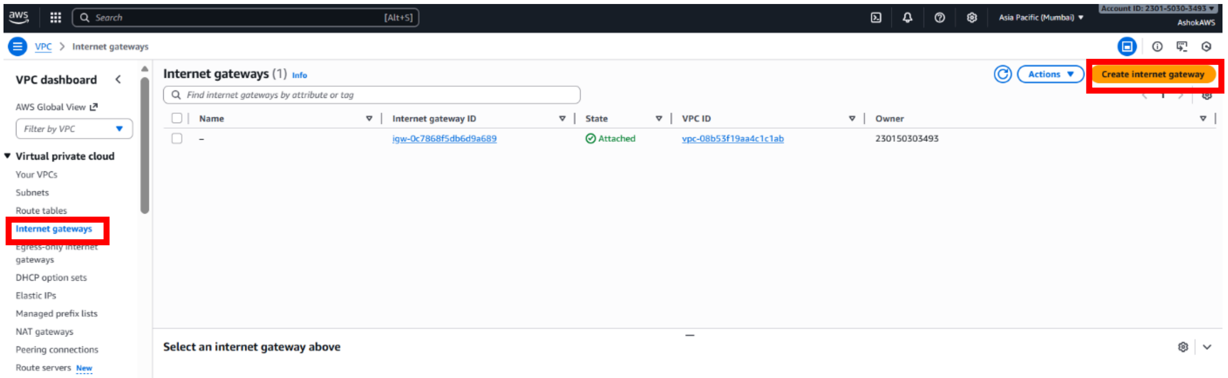


A success message will be displayed **after creating subnet (PrivateSubnet).**

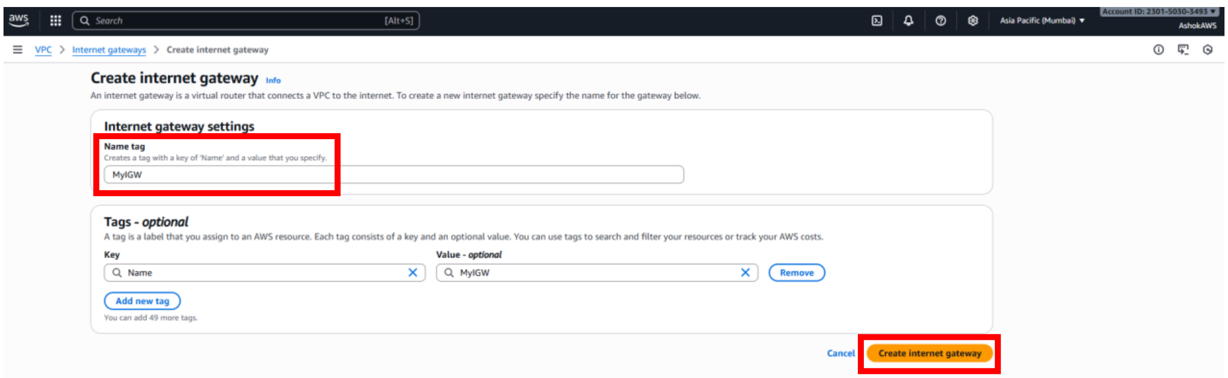


4	**Create an Internet Gateway**

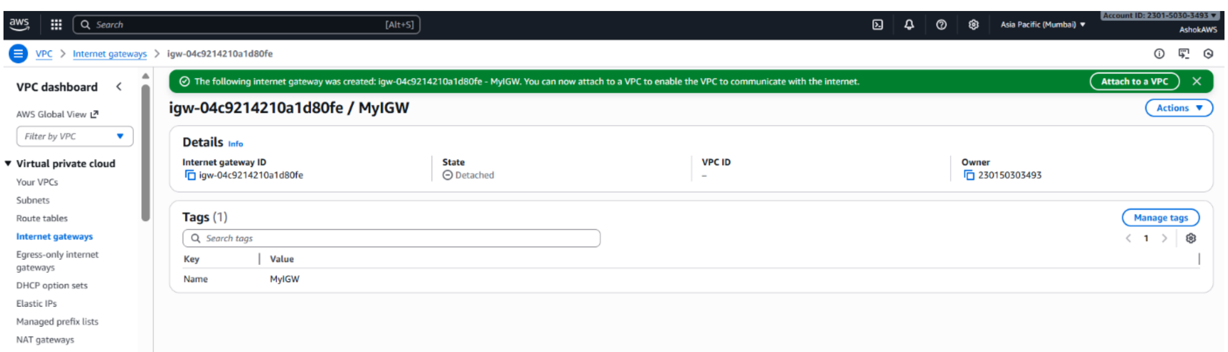
Step 4.1 Click **VPC → Internet gateways → Create internet gateway**



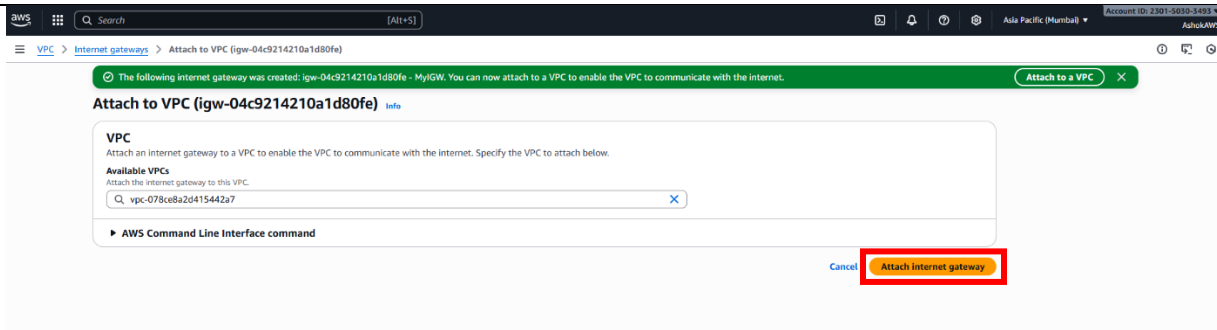
Add the detail shown below, then click **"Create internet gateway" button**



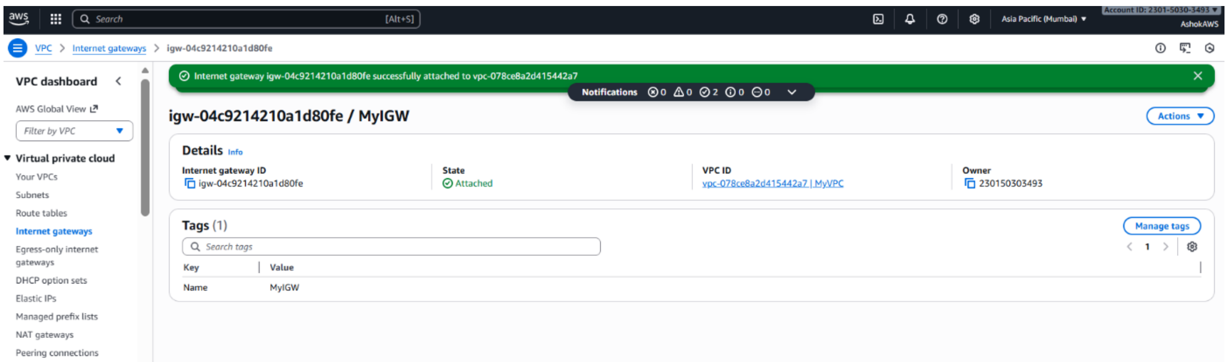
A **message** will be displayed after creating internet gateway.



Step 4.2 Once the internet gateway is created, attach it to the VPC using the **"Attach internet gateway" button.**

<table>
<tr>
<td></td>
<td>


A success message will be displayed **after the internet gateway is attached to the VPC.**

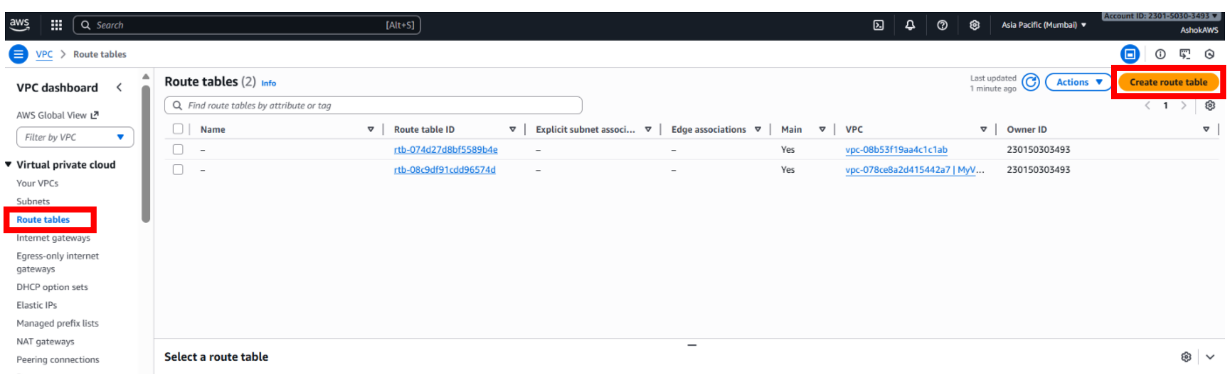


</td>
</tr>
<tr>
<td>5</td>
<td>

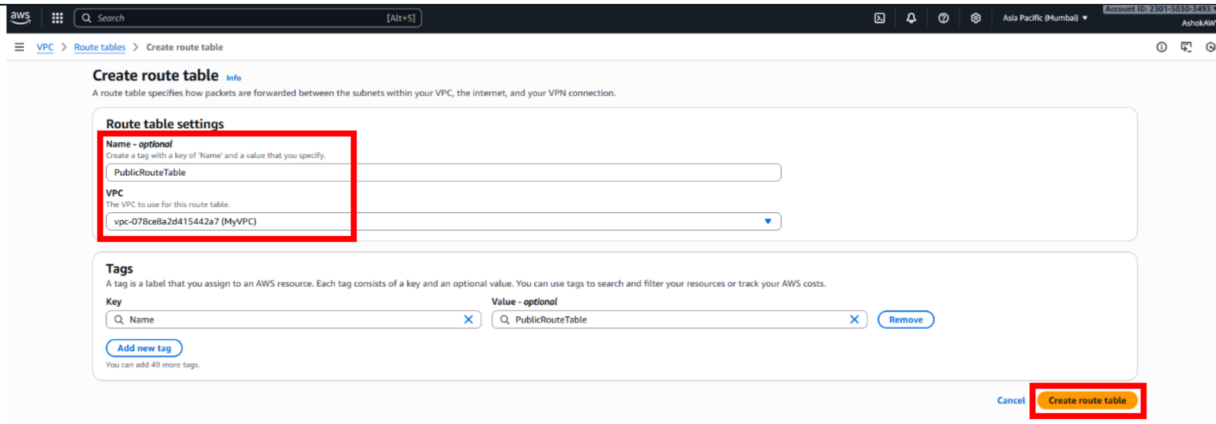
Configure Route Tables

Create Public Route Table

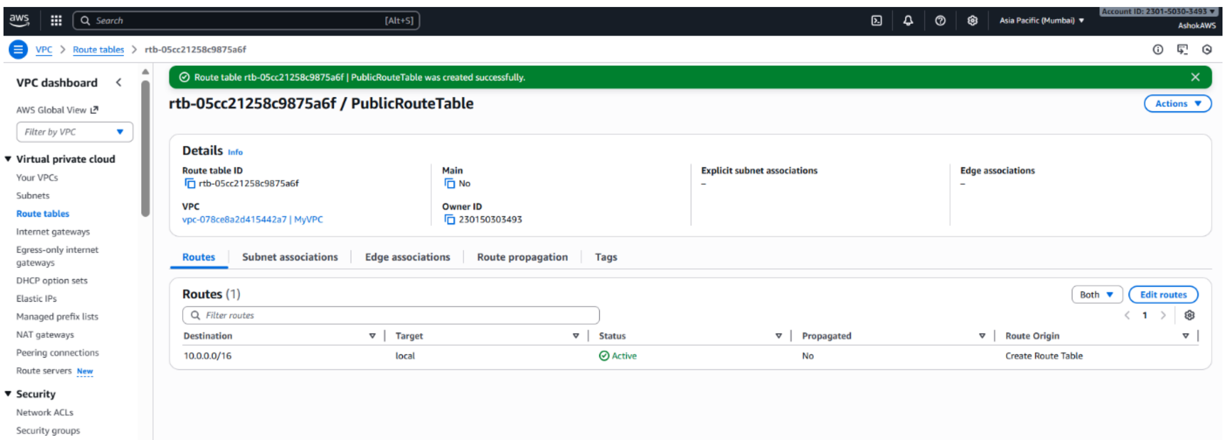
Step 5.1 Click **VPC → Route tables → Create route table**



Add the details shown below, then click **“Create route table” button**
</td>
</tr>
</table>



A success message will be displayed **after creating route table (PublicRouteTable).**



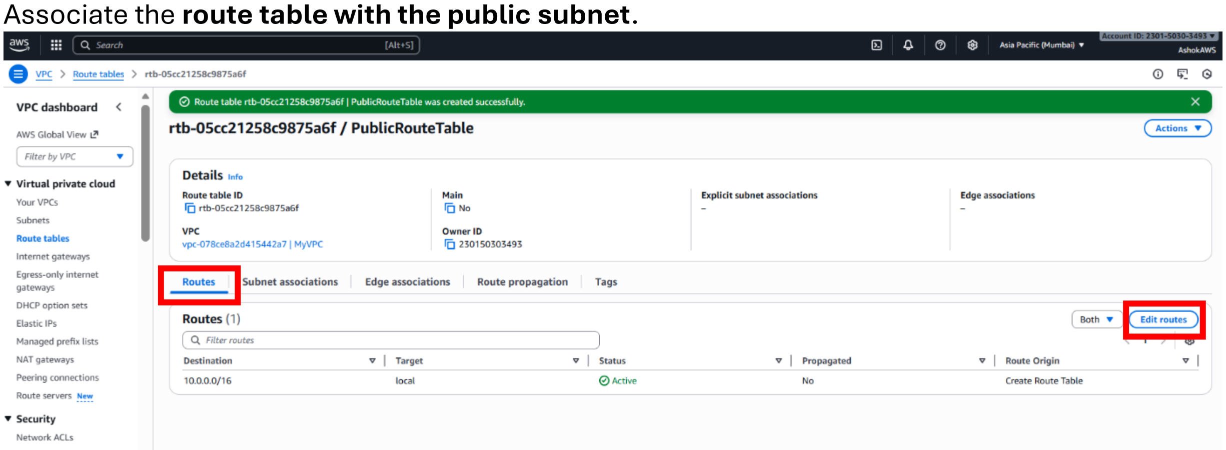
Step 5.2 In Routes section, click **"Edit routes" button**, then click **"Add route" button**

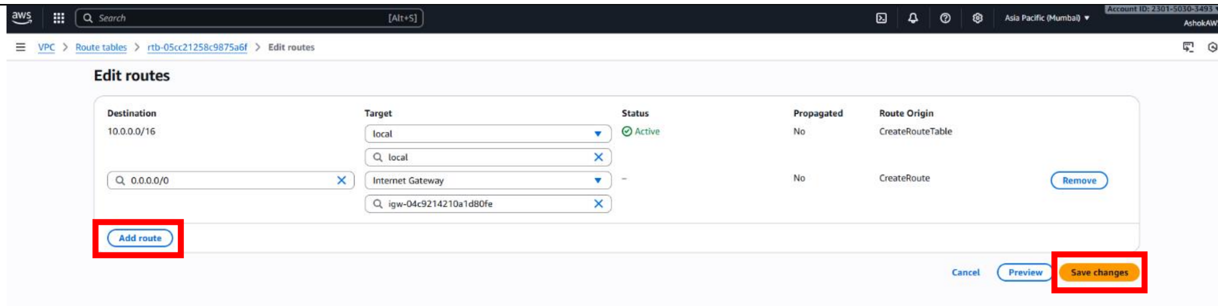
Add the details:

- **Destination: 0.0.0.0/0**
- **Target: select your Internet Gateway**

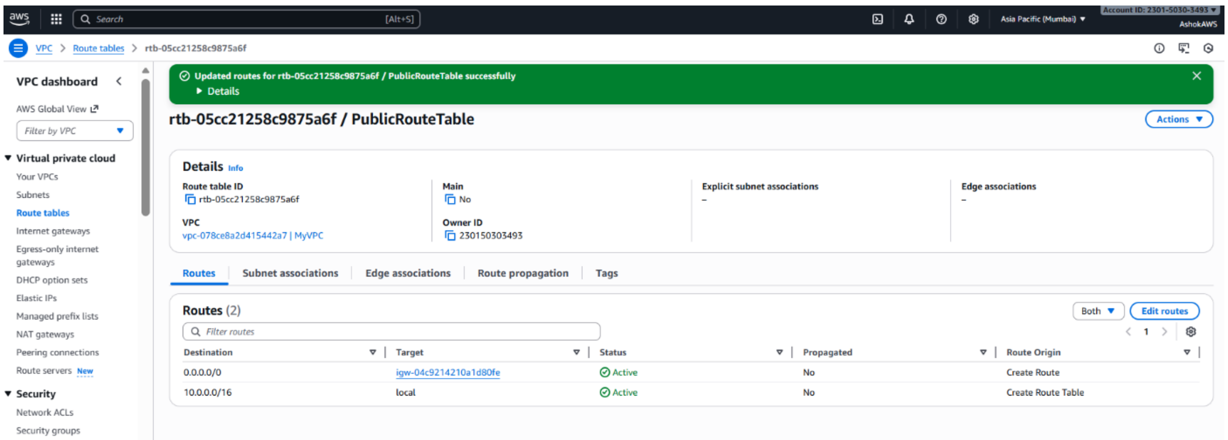
then click **"Save changes" button**

Associate the **route table with the public subnet**.



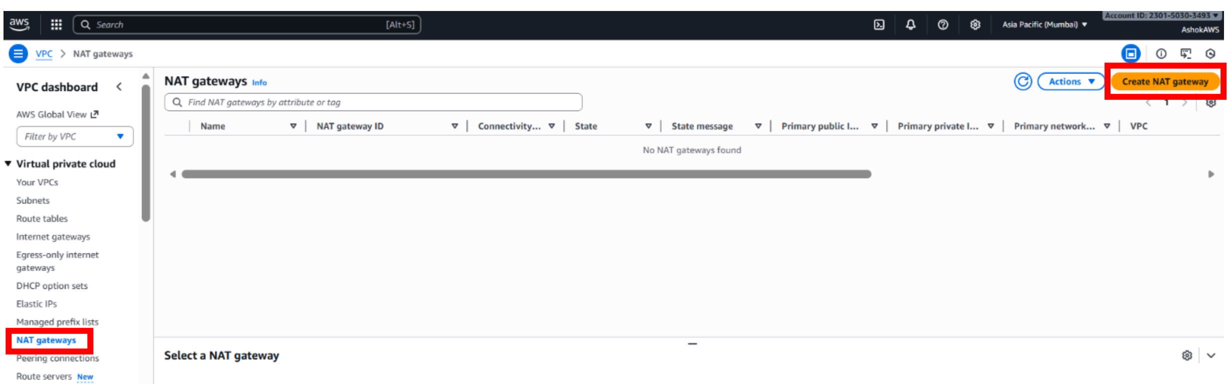


A success message will be displayed **after the route is updated for the public route table.**



Create Private Route Table

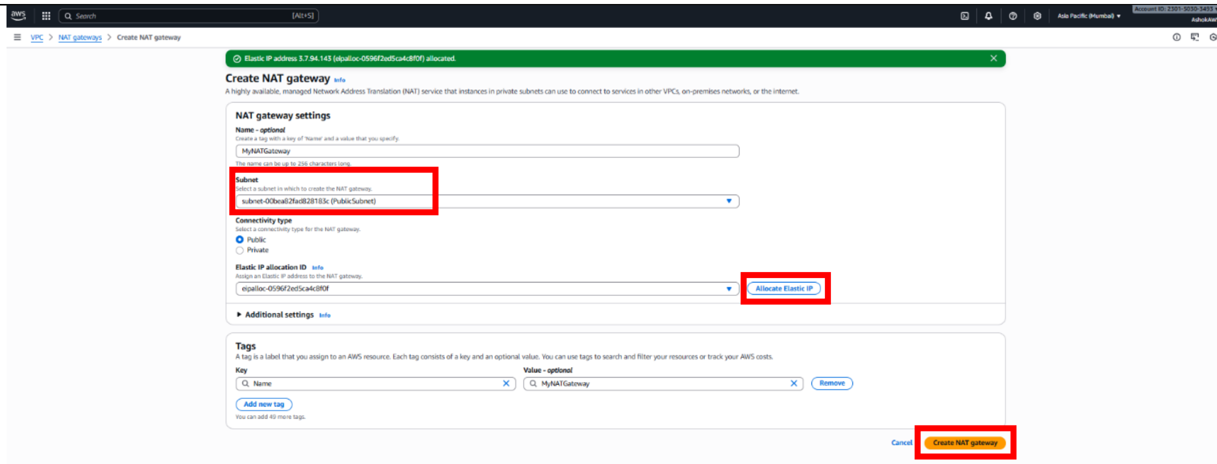
Step 5.3 Click **VPC → NAT gateways → Create NAT gateway**



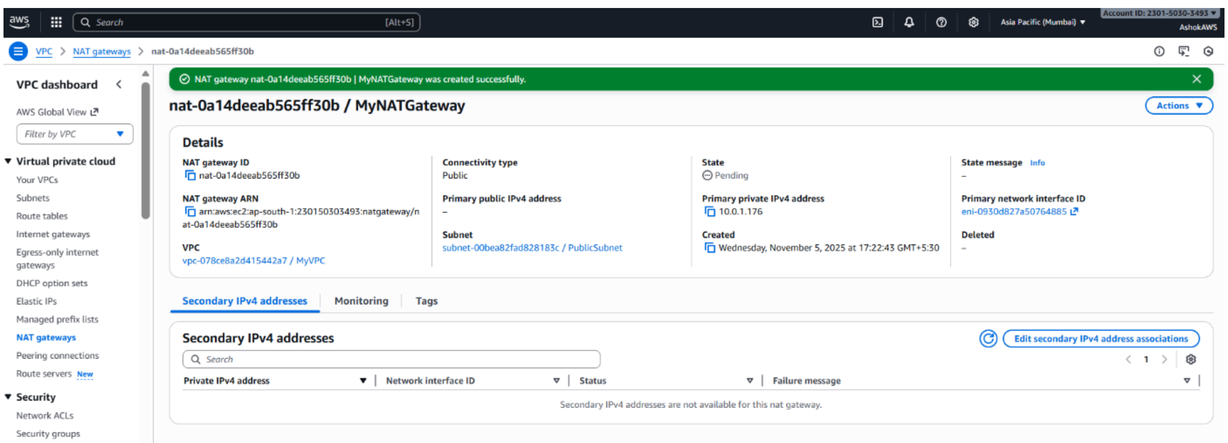
Add the details:

- **Subnet: select your public subnet**
- **Elastic IP allocation ID:** Click **"Allocate Elastic IP"**

then click **"Create NAT Gateway" button**



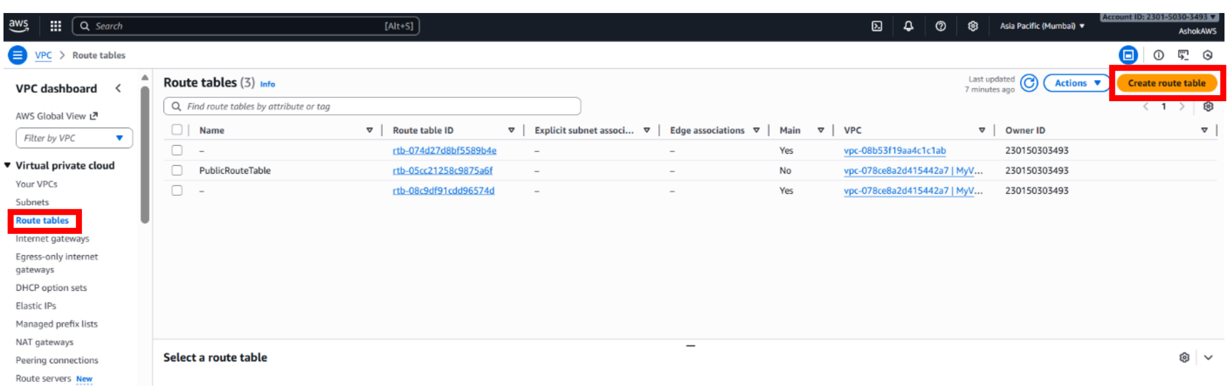
A success message will be displayed **after creating NAT gateway.**



Update the Private Route Table

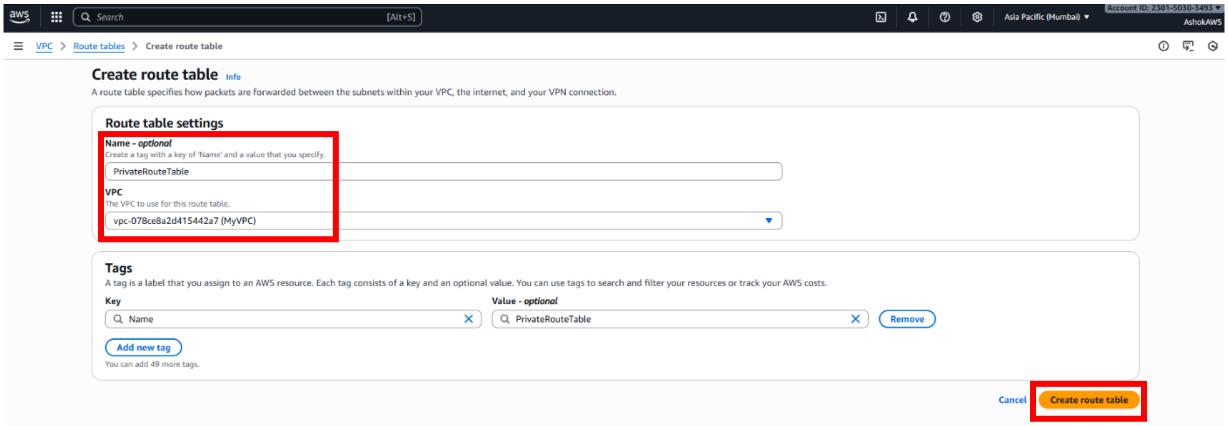
Now, **configure the private subnet to use the NAT Gateway for internet traffic.**

Step 5.4 Click **VPC → Route tables → Create route table**

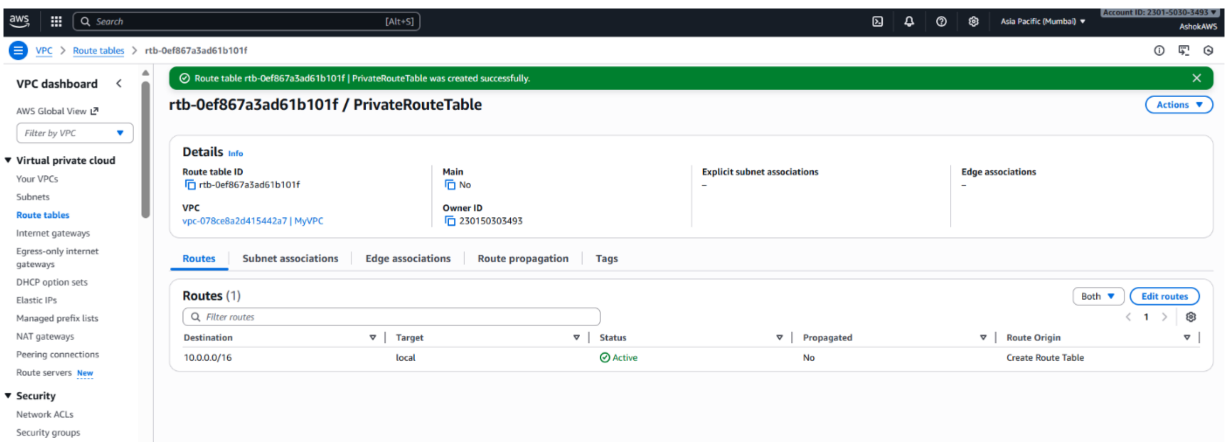


Step 5.5 If you don't have a PrivateRouteTable yet, create one:
Add the details:

- **Name:** PrivateRouteTable
- **VPC:** select your VPC
- Click **Create route table**

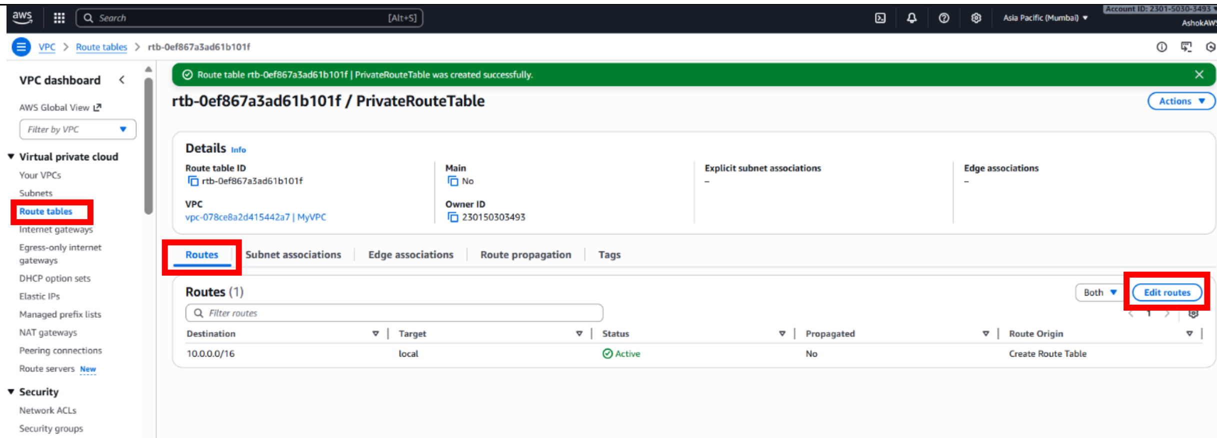


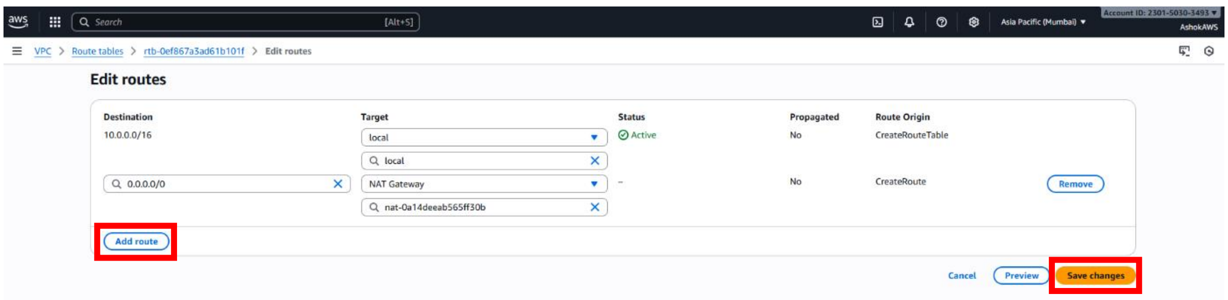
A success message will be displayed **after creating route table (PrivateRouteTable).**



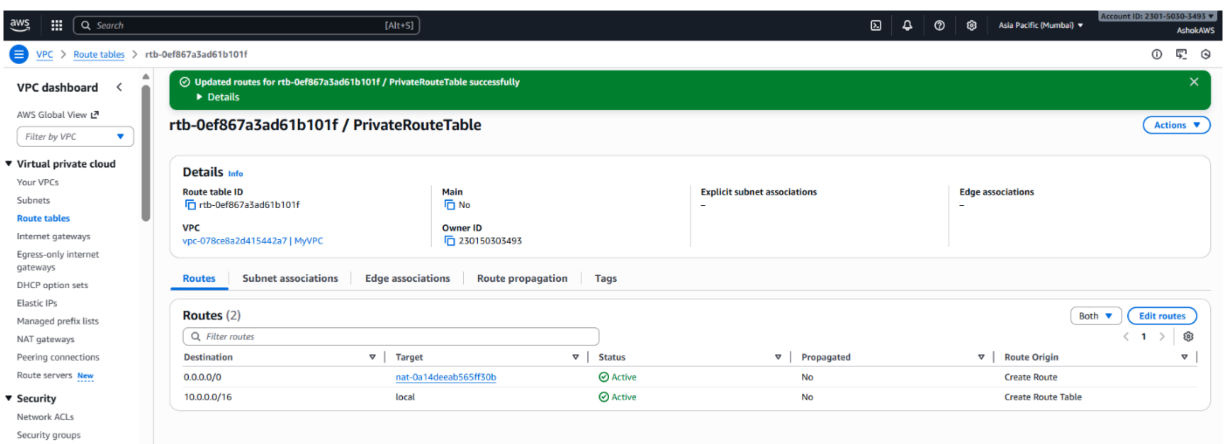
Step 5.6 Click **Route tables** → go to **Routes tab** → click **Edit routes** → click **Add route**
Add the details:

- **Destination:** 0.0.0.0/0
- **Target:** your new NAT Gateway (nat-xxxxxxxx)
- Click **Save changes**

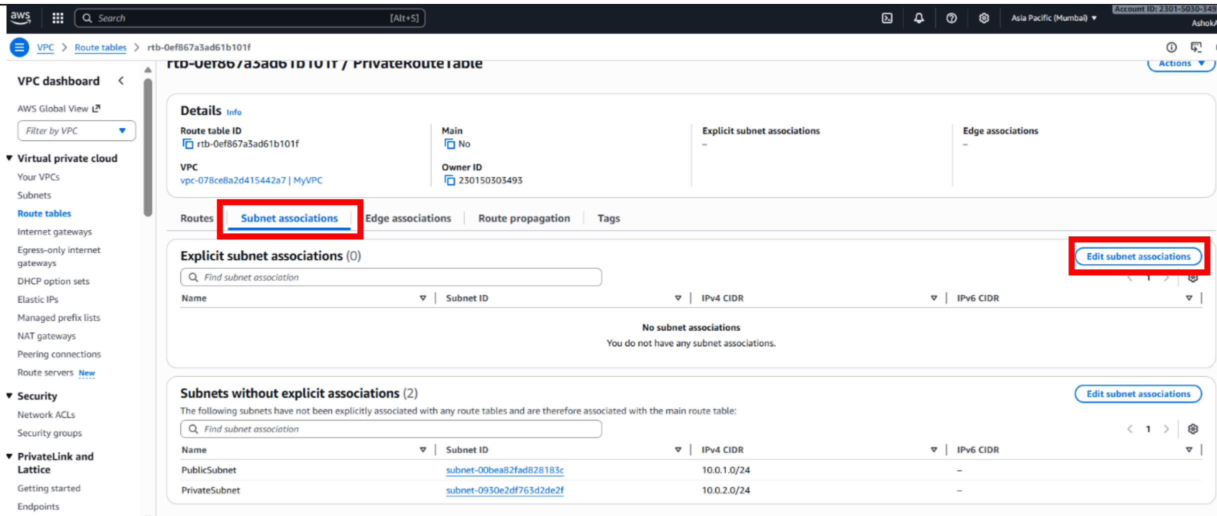




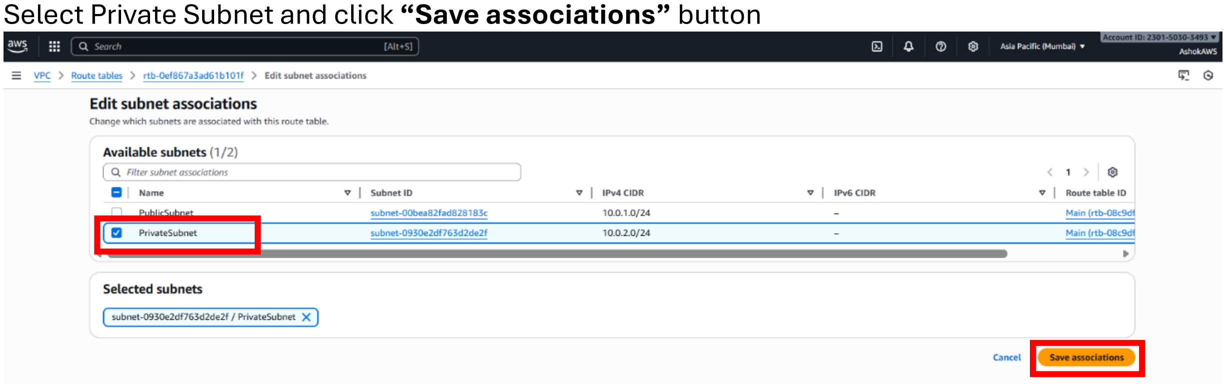
A **success message** will be displayed **after the route is updated for the private route table.**



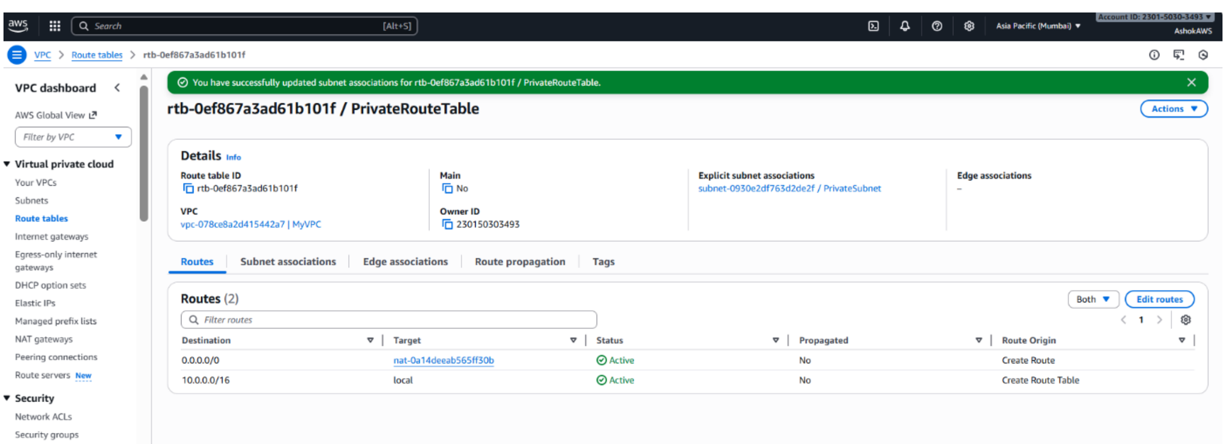
Step 5.7 Go to **Subnet associations tab** → click **"Edit subnet associations" button**



Select Private Subnet and click **“Save associations”** button



A success message will be displayed **after the subnet associations is updated for the private route table.**



Thus, the **VPC has been created successfully with public and private subnets**.